
The Two Roles of the Physics Residual in Neural CFD Surrogates: A Calibrated Trust Signal but a Poor Correction Objective

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Abstract

Machine-learning surrogates for computational fluid dynamics (CFD) predict steady flow fields far faster than classical solvers, but emit a single field with no built-in way to know whether to trust it—especially out of distribution. We close the loop with the governing physics: we compute the discretised steady-state Reynolds-averaged (RANS) residual of the prediction and ask what jobs it can do. Our central finding is a clean **two-way dissociation**: the physics residual is a reliable, backbone-robust **trust signal** (it tells you *where* the prediction is wrong) but a poor **correction objective** (it does not tell you *how* to fix it). *As a trust signal*, the residual’s per-case rank correlation with field error is consistently positive across three architecturally distinct backbones and a second dataset of laminar bluff bodies; it flags the worst-decile cases with AUROC ≈ 0.9 , and a distribution-free split-conformal layer attains its target coverage. *As a correction objective or acceptance gate*, the residual fails: reducing it does not reduce field error, and a backtracking test admits almost no residual-reducing step. Separately, a learned deep-equilibrium corrector trained toward ground truth lowers volume-field error on a state-of-the-art backbone; a controlled ablation that removes the residual input matches this gain, so the improvement comes from the learned correction, not from conditioning on the residual. The contribution is a self-auditing, calibrated trust layer, the residual’s two roles (trust signal yes, correction objective no), and the learned correction it accompanies, demonstrated on a competitive surrogate and released as the open-source **neuroforge-cfd** package.

Keywords: neural operators, surrogate CFD, physics residuals, uncertainty quantification, conformal prediction, trust calibration, airfoil aerodynamics.

1 Introduction

Classical CFD resolves the governing PDEs by iterating a discretised system until its residuals fall below a tolerance. It is accurate and trusted, but slow: meshing and a converged steady-RANS solve over a new geometry can take minutes to hours, throttling the rapid design-space exploration that early engineering most needs. Neural surrogates promise the opposite trade—millisecond inference—and a vigorous literature now maps geometry plus boundary conditions directly to flow fields with neural operators and attention models.

The problem is **trust**. A trained surrogate is a one-shot function: it emits a field with no self-assessment, no convergence signal, and no recovery path when it extrapolates. On a geometry far from the training set—the common case in design—the user cannot distinguish a faithful prediction from a confident hallucination.

The governing physics offers an obvious lever. The steady incompressible RANS residual of any candidate field is computable directly from the field, with no ground truth. A neural-CFD pipeline could use this residual in two distinct ways: as a *trust signal* that flags *which* predictions to distrust (a case-level error proxy that drives a conformal trust layer), and as a *correction objective or acceptance gate* that a step must reduce. These are not the same role, and—this is the crux—they do not earn the same verdict. Put plainly, the residual tells you *where* a prediction is wrong, not *how* to fix it. This paper separates the two roles and answers each empirically on a SOTA backbone, and—separately—deploys a learned, supervised corrector whose gain we attribute by ablation to the learned correction, not to the residual.

Research question. *In which of its two roles—trust signal and correction objective/gate—is the steady-RANS physics residual of a neural-CFD prediction useful, and does the answer survive on a competitive backbone?*

Thesis (one sentence). *The steady-RANS physics residual is a reliable, conformally-calibrated, backbone-robust case-level trust signal—it tells you where a prediction is wrong—even though minimising the residual directly remains a poor correction objective; the learned correction we deploy reduces error on a SOTA surrogate, but a controlled ablation attributes that gain to the learned correction, not to feeding it the residual.*

We arrive at this thesis the hard way. We built the full predict→verify→estimate-uncertainty→correct→fall-back pipeline, pre-registered hypotheses, and ran a 3-seed ablation on real AirfRANS plus an out-of-distribution study, two formal certificates, and—in a second phase—the same trust layer and corrector on a SOTA Transolver backbone. The trust signal and the supervised corrector survive on the strong backbone; the residual-as-objective machinery (the iteration sweep and the acceptance gate) is the part that disappoints, and we report that as a finding rather than hide it. Two honest findings accompany the corrector: *correction quality is backbone-dependent*—the supervised corrector that helps on the SOTA backbone is flat on the weak grid backbone—and a controlled ablation shows the gain is *not* sourced from the residual input (a null corrector with the residual channels zeroed matches or slightly exceeds it), so we attribute the gain to the learned correction, not to residual-conditioning. We report both.

1.1 Contributions

In priority order:

1. **A backbone-agnostic trust layer over a consistently positive residual detector (headline).** The physics residual is a calibrated, *case-level* error detector across three architecturally-distinct backbones tested. On the SOTA Transolver backbone its per-case rank (Spearman) correlation with field error is 0.625 ± 0.019 (subsection 5.2, Table 1); on a much weaker grid backbone it is ≈ 0.40 bare and a learned corrector lifts it to 0.83 (subsection 5.3); and on a param-matched message-passing GNN (MeshGraphNet, bare) it is 0.851 ± 0.058 (results/mgn/mgn_results.json, a high value that partly reflects that model’s wide error spread, which is easy to rank), so the detector is consistently *positive* across all three, though its *strength* varies (ρ from 0.40 to 0.85; per-cell spatial localisation is weaker still, section 3). Building on it—and depending only on a model’s predictions and σ , hence backbone-agnostic—a split-conformal trust layer attains target coverage on the SOTA backbone (0.902 ± 0.008 at the 0.90 target, subsection 5.2); with a deep-ensemble σ the band is input-adaptive on that same near-deterministic model (ECE 0.074, Table 2), and on the weak backbone it holds under distribution shift, including a regime where the bare model’s signal collapses (subsection 5.4, subsection 5.6). The signal also supports *selective prediction*: it detects worst-decile-error cases with AUROC 0.87–0.91 on the AirfRANS arms (0.83–0.98 including the DeepCFD OOD seeds), and rank-fused with the ensemble σ reaches AUROC 0.905, recovering $\approx 91\%$ of the oracle’s achievable error reduction at a 10% rejection budget (Figure 2). The layer depends only on a model’s predictions and uncertainty, so it is backbone-robust.
2. **Learned correction works on a strong backbone.** A learned DEQ corrector trained (supervised) toward ground truth reduces volume-field MSE on the SOTA backbone on **all three seeds**:

mse_u 0.127 $\rightarrow$ 0.116 (−9%), mse_v 0.100 $\rightarrow$ 0.079 (−21%), mse_p 630 $\rightarrow$ 471 (−25%, gate-verified), with internal force-ranking self-consistency held ($\rho_{C_l} = 0.999$; against *official* labels drag ranking is recovered at $\rho_D = 0.84$, and a repaired control-volume integrator recovers official *lift* nearly exactly from the predicted fields, $\rho_L = 0.998 \pm 0.001$ with per-seed median magnitude errors of 3.6–3.9%, subsection 5.2). The backbone is a published SOTA architecture (Transolver), so the corrector improves a competitive surrogate. The deployed corrector ingests the residual as an input channel, but a controlled ablation that zeros that input at train and inference (3 seeds) matches or slightly exceeds the deployed corrector, so we attribute the gain to the learned correction, not to residual-conditioning (subsection 5.2, results/control/w1_capture.json).

3. **Residual-as-objective fails (the surviving negative).** Used as a correction *objective or acceptance gate*, the residual does not work. Sweeping the correction iterations *raises* the PDE residual (0.11 $\rightarrow$ 0.62) while *lowering* field error (mse_u 3.92 $\rightarrow$ 2.29): the two move in opposite directions (subsection 5.5, Figure 6). A backtracking acceptance test that admits only residual-reducing steps accepts essentially zero steps on a trained corrector (subsection 5.5). Minimising the residual is not minimising the error—which is exactly why the deployed corrector supervises toward truth rather than minimising the residual.
4. **Backbone-dependence, reported as a finding (not hidden).** The same corrector that delivers −9 to −25% on the SOTA backbone is flat-to-unhelpful on the weak grid backbone (subsection 5.3, Table 3), where it trades volume accuracy for surface fidelity and a stronger trust signal. Correction quality is therefore backbone-dependent; we report both regimes side by side and the SOTA result is the one that supports the learned-correction claim. For honest context we also report a matched-budget baseline study (subsection 5.7, Table 6): the grid backbone trails Transolver by 4–60×, while the trust layer and corrector are demonstrated *on* the competitive Transolver backbone in subsection 5.2.
5. **A reproducible, CPU-first open-source package (neuroforge-cfd)** with a frozen I/O contract, interchangeable backbones, a zero-download synthetic potential-flow data generator, an AirfRANS loader, the ablation/certificate harness, and the figures—so every claim runs on a laptop and every certificate is reproducible from a committed script.

We are explicit throughout about what is **measured** versus **assumed**, retain the retraction of stale single-seed numbers from earlier drafts (subsection 5.3), and make no out-of-distribution coverage *guarantee*; we make a competitiveness claim only for the Transolver backbone our trust layer runs on (subsection 5.2), not for the grid backbone.

2 Related Work

Neural operators (FNO / Geo-FNO). Fourier Neural Operators learn a resolution-agnostic mapping between function spaces by parameterising a global convolution in the spectral domain (Li et al., 2021). Geo-FNO (Li et al., 2022) extends this to irregular geometries via a learned deformation to a latent uniform grid where the FFT is valid. These are strong *one-shot* operators with no self-assessment; NeuroForge implements both as backbones and adds the trust layer around them.

Point-cloud and physics-attention operators (DoMINO, Transolver). DoMINO (Ranade et al., 2025) is a decomposable multi-scale point-cloud operator for external aerodynamics on large industrial meshes. Transolver (Wu et al., 2024) replaces quadratic attention with attention over learnable *physics slices*, giving linear cost and strong accuracy on PDE benchmarks; Transolver++ (Luo et al., 2025) scales it to million-scale meshes. We use Transolver in two ways: as a matched-budget baseline for our grid backbone (subsection 5.7), and as the SOTA backbone our trust layer and corrector actually run on (subsection 5.2), where the residual is a calibrated detector and a supervised learned corrector reduces field MSE.

Learned solver-correctors and the “fixer” premise. Several lines of work use the discretised residual or a coarse-solution error as a *correction objective*: learned PDE solvers with convergence guarantees (Hsieh

et al., 2019), deep-equilibrium operators for steady PDEs (Marwah et al., 2023), iterative refiners (Lippe et al., 2023), and learned residual-error correctors (Jha, 2024). PINNs (Raissi et al., 2019) embed the residual in the training loss. NeuroForge implements a contractive DEQ corrector and a feed-forward residual-conditioned corrector squarely in this family. **Our contribution to this line is to test the residual-as-correction premise head-on and find it does not hold.** As a correction *objective*—where a step must reduce the residual—residual minimisation does not track error minimisation on a real RANS benchmark (the residual rises as error falls; the acceptance gate accepts almost nothing). And when we feed the residual as an input *feature* to a corrector supervised toward ground truth, a controlled ablation shows the residual input is not the source of the gain: zeroing it at train and inference (3 seeds) matches or slightly exceeds the deployed corrector (subsection 5.2, results/control/w1_capture.json). The residual’s value is as a *trust signal*—it tells you *where* a prediction is wrong—not as a correction mechanism telling you *how* to fix it; the learned correction does the fixing.

Residual-based, data-free error signals. Closest to our *trust-signal* half is recent work that reads the PDE residual as a reference-free indicator of prediction error. Roy et al. (2025) use a physics-informed, residual-based error estimator to detect and correct accumulating error in neural-operator *time-marching*, triggering a classical solver when the estimator fires; Song et al. (2026) design an epistemic-UQ scheme whose uncertainty bands are aligned with localised residual structure in operator surrogates. Gopakumar et al. (2025) go furthest in the trust direction: they use the PDE residual itself as a conformal nonconformity score, obtaining data-free coverage guarantees—in *residual* space, with the transfer to solution space left open as a set-propagation problem. Our residual-floor theorem (section 6) supplies a concrete mechanism for that gap in the discrete steady-RANS setting: the monitored residual has a nonzero floor at the reference solution and a kernel of undetectable modes, so residual-space smallness cannot by itself certify solution-space accuracy. From the theory side, Mukherjee et al. (2026) prove that vanishing residuals guarantee solution convergence only under additional compactness assumptions; our setting is the practical complement, where the discrete monitored residual provably does *not* vanish at the dataset truth. We share the premise that the residual is a data-free error signal and do not claim to originate it. Our distinct contributions are (i) the head-to-head *dissociation*—the residual is a good trust signal but a poor correction objective, established on the same steady-state RANS benchmark; (ii) the *residual-floor theorem* quantifying the operator-specific detection limit and the undetectable kernel modes; and (iii) a split-conformal certificate calibrated on the *deployed, corrected* field. Our setting is steady-state external aerodynamics with per-case spatial trust, rather than time-marching error control.

Uncertainty quantification and conformal prediction for PDE surrogates. Deep ensembles (Lakshminarayanan et al., 2017) and MC-dropout (Gal & Ghahramani, 2016) are standard epistemic estimators; conformal prediction gives distribution-free coverage and has been applied to operator learning (Ma et al., 2024). Concurrently with this work, Jia et al. (2026) develop multi-granularity conformal calibration (case-level quantile regression plus residual-normalised point-level bands) for neural-operator automotive surrogates on DrivAerML; their nonconformity scales are purely data-driven, whereas our trust signal is the physics residual itself, and we additionally test—and reject—its use as a correction objective. We do **not** propose a new UQ method. Our trust layer applies split-conformal calibration to MC-dropout uncertainty; the new content is the *characterisation* of when this holds for CFD surrogates—in particular that coverage survives out-of-distribution empirically via uncertainty inflation, while the formal guarantee does not transfer once exchangeability fails.

Benchmarks (AirfRANS). AirfRANS (Bonnet et al., 2022) provides ~ 1000 incompressible steady-RANS simulations over NACA airfoils with **full**, **scarce**, **reynolds**, and **aoa** splits designed to probe generalisation. We use **full** for in-distribution evaluation and the regime-disjoint **reynolds/aoa** splits for the out-of-distribution study.

2.1 Positioning

The closest prior art treats the physics residual primarily as a *correction objective* (PINNs, FNO-DEQ, learned correctors) or treats UQ in isolation from physics. NeuroForge’s distinctive question is which of

two roles the residual can play, evaluated head-to-head on the same model and data and then on a SOTA backbone: as a trust *signal* (detect error, drive conformal calibration) and as a correction *objective/gate*. Our answer—signal yes (it tells you *where* the prediction is wrong), objective no—and the resulting calibrated trust layer are the contribution. Separately, a learned, supervised corrector reduces field error on the strong backbone, but a controlled ablation attributes that gain to the learned correction rather than to the residual input. The framework’s no-harm/contraction machinery (section 3) is retained because it is correct and reproducible; the acceptance-gate guarantee in particular is real but, as we show, vacuous as an accuracy mechanism.

3 Method

3.1 Pipeline overview

The NeuroForge framework wraps a one-shot backbone in a verify–estimate–correct–fall-back loop (Figure 1). Throughout, the physics residual plays **two distinct roles with two distinct verdicts** (section 5): (i) as a *trust signal* it detects error and drives the conformal layer (reliable, backbone-robust—it tells you *where* the prediction is wrong); (ii) as a correction *objective or acceptance gate* it fails (subsection 5.5). Separately, a learned, supervised corrector improves a SOTA backbone (subsection 5.2); the deployed corrector ingests the residual as an input channel, but a controlled ablation attributes its gain to the learned correction, not to the residual input. We describe all components for completeness and flag, at each one, which role it instantiates.

3.2 Geometry encoding and governing equations

A **FlowCase** (geometry + boundary conditions + fluid + domain) is encoded into a fixed channel-first stack $x \in \mathbb{R}^{7 \times H \times W}$ in the frozen `INPUT_CHANNELS` order (`sdf`, `mask`, $x, y, u_{\text{in}}, v_{\text{in}}, \log \text{Re}$): the signed distance to the body surface (negative inside the solid), a fluid/solid mask, normalised cell coordinates, the freestream velocity broadcast over the grid, and $\log_{10} \text{Re}$. The backbone outputs `OUTPUT_CHANNELS` (u, v, p, ν_t), where p is the **kinematic** pressure p/ρ .

The verifier evaluates the steady, incompressible, 2-D RANS equations in primitive form on the **physical (denormalised)** fields, with effective viscosity $\nu_{\text{eff}} = \nu + \nu_t$, pointwise:

$$\text{continuity: } r_c = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}, \quad (1)$$

$$x\text{-momentum: } r_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} - \nu_{\text{eff}} \nabla^2 u, \quad (2)$$

$$y\text{-momentum: } r_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} - \nu_{\text{eff}} \nabla^2 v. \quad (3)$$

Because p is kinematic, no explicit density appears. Derivatives use central finite differences with one-sided stencils at the borders; residuals are zeroed inside the solid. A boundary-condition violation map r_{bc} adds a **no-slip** term penalising velocity magnitude in the thin fluid band adjacent to the wall (weighted by $\exp(-|\text{sdf}|/\ell)$, $\ell \approx 3$ cells) and a **far-field** term penalising deviation from (u_∞, v_∞) on the outer border ring. We stress the standard caveat that drives our whole study: a low residual is **necessary but not sufficient** for correctness—a smooth near-freestream field can have near-zero residual yet be entirely wrong—so the residual is a *consistency monitor*, and whether it tracks error is the empirical question section 5 answers.

3.3 Trust map and conformal calibration (the surviving contribution)

Raw trust map. The combined PDE-residual magnitude $\rho_{\text{res}} = \sqrt{r_c^2 + r_x^2 + r_y^2 + r_{bc}^2}$ and a per-cell predictive uncertainty σ are each mapped to $[0, 1]$ (using an absolute physical reference scale $s = U_\infty^2/L + U_\infty/L$ when available, else a robust 95th-percentile normalisation with an absolute floor) and fused,

$$e = \text{clip}(w_r \hat{r} + w_u \hat{\sigma}, 0, 1), \quad T = 1 - e, \quad (4)$$

Split-conformal calibration. Raw σ is uncalibrated, so a threshold on it carries no guarantee. We add split-conformal calibration (cf. Ma et al., 2024): on a held-out calibration set we compute per-channel nonconformity scores $s = |\hat{y} - y|/\sigma$ and take the finite-sample-corrected $(1 - \alpha)$ quantile q ; the band $q \cdot \sigma$ then satisfies the distribution-free coverage guarantee in Proposition 1.

Proposition 1 (Distribution-free coverage under exchangeability). *If the calibration scores and a test score are exchangeable, then the split-conformal band $q \cdot \sigma$ satisfies $\mathbb{P}(|\hat{y} - y| \leq q \sigma) \geq 1 - \alpha$.*

This converts the trust threshold from a hand-picked constant into a band with a known in-distribution coverage level. subsection 5.6 reports the measured coverage, the coverage-vs- α sweep, and the out-of-distribution behaviour, with the exchangeability caveat made explicit.

3.4 The correction loop: supervised corrector vs. residual objective/gate

The framework offers two correction operators and a backtracking acceptance test, and the distinction between them *is* the intellectual core of the paper. The DEQ corrector is trained by supervised regression toward ground truth and applied directly without any acceptance test; it ingests the residual as an input channel (its deployed architecture, below), but its gain is attributable to the learned correction rather than to that input (the W1 ablation, subsection 5.2). This is the path that helps on a SOTA backbone (subsection 5.2). The backtracking acceptance test uses the residual as an *objective/gate*—a step is admitted only if it lowers the residual norm; this is the path that fails (subsection 5.5). We describe both precisely so the contrast is interpretable.

Feed-forward corrector + backtracking acceptance test. A small residual CNN c_ϕ predicts an additive correction conditioned on the current field, its current 3-channel residual map, and the geometry: $\Delta_k = c_\phi(y_k, R(y_k), x)$. A candidate $y_{k+1} = y_k + s \Delta_k$ is accepted only if it does not increase the residual norm $N(y) = \sqrt{r_c^2 + r_x^2 + r_y^2}$.

$$N(y_k + s \Delta_k) \leq N(y_k) + \varepsilon, \quad (5)$$

with s halved up to four times on failure. By construction $N(y_0) \geq N(y_1) \geq \dots \geq N(y_K)$ —the residual norm is monotone non-increasing across accepted steps. This is the convergence discipline of a classical solver. Its weakness is exactly the one we exploit as a finding: the test is satisfiable by the identity (zero step), so a corrector that cannot reduce the residual is simply rejected. Empirically (subsection 5.5) a trained corrector accepts essentially no steps, because reducing the field error and reducing the PDE residual are not the same objective.

Contractive DEQ corrector. To give the loop a genuine convergence guarantee in its *own* iteration variable, the correction δ is defined as the fixed point of a learned operator

$$\delta^* = T_\theta(\delta^*; c), \quad T_\theta(\delta; c) = \kappa \cdot g_\theta([\delta, c]), \quad c = [\hat{y}, r(\hat{y}), \text{geom}], \quad (6)$$

where g_θ is a CNN whose layers are spectrally normalised (each ≤ 1 -Lipschitz) and $\kappa < 1$.

Proposition 2 (Banach contraction of the corrector). *With g_θ each-layer 1-Lipschitz and $\kappa < 1$, T_θ is a κ -contraction in δ . By the Banach fixed-point theorem the equilibrium δ^* exists, is unique, and the iteration converges geometrically, $\|\delta_k - \delta^*\| \leq \kappa^k \|\delta_0 - \delta^*\|$.*

The construction underlying Proposition 2 is a Deep-Equilibrium model (Bai et al., 2019) with the Lipschitz constant controlled by spectral normalisation (Winston & Kolter, 2020), trained with Jacobian-Free Back-propagation (Fung et al., 2022). Crucially, g_θ is trained on **data** (it targets the correction toward truth, with the residual as an input *feature*) rather than by minimising the residual; at inference the converged δ^* is applied directly, **without** the backtracking acceptance test. We verify this contraction empirically (subsection 5.6: measured factor $0.78 < \kappa = 0.9$). Two scopes must be kept distinct, and subsection 5.5 leans on the distinction: the DEQ contraction is in the *correction variable* δ , whereas the *PDE residual of the output field* is a different quantity that, as we show, can rise even as δ converges and the field error falls.

The feed-forward corrector is trained (backbone frozen) so that $c_\phi(\hat{y}_{\text{norm}}, R_{\text{norm}}, x) \approx y_{\text{norm}}^* - \hat{y}_{\text{norm}}$, with the final convolution initialised near zero for a stable first iteration.

3.5 Uncertainty-gated classical fallback

After the loop, if the maximum uncertainty exceeds a threshold, the low-trust fluid region is handed to a `ClassicalFallback`. In the present release this is an **interface with a stub backend** that reports what would run (and the region size) without invoking an external solver; the `openfoam/su2` backends raise `NotImplementedError` with setup guidance. This keeps the package importable and runnable with nothing installed while fixing the integration seam; it is not part of any quantitative claim.

4 Implementation

NeuroForge is an open-source, **CPU-first**, pure-Python package (`neuroforge-cfd`, Python ≥ 3.10 , NumPy/SciPy/PyTorch/Matplotlib). Importing the package caps BLAS/OMP/MKL thread counts to one by default to avoid catastrophic oversubscription on low-core machines, and does no heavy work. The codebase is organised around a **frozen I/O contract** in `core/`.

Module map. `core/` holds the frozen data contracts (`FlowCase`, `FlowField`, `Diagnostics`, `SolveResult`) and `Config`. `geometry/` builds the 7-channel input (NACA generation, SDF/mask rasterisation, `encode_case`). `data/` has the synthetic generator, the AirfRANS loader, the rasteriser, and the datamodule (`Normalizer/loaders`). `models/` provides `FN02d`, `GeoFNO`, a Transolver-style `PhysicsTransformer`, `UNet/DeepONet` baselines, `LocalCorrectionNet`, the `DEQCorrector`, and the `DeepEnsemble/MCDropoutUQ` wrappers, all in a string-keyed registry. `physics/` has the differential operators, the `PhysicsChecker`, `trust_map`, force/Cp/error metrics, the conformal calibration, and the differentiable `physics_residual_torch`. `solver/` has `Predictor`, `NeuroForgeEngine`, `neural_residual_iteration`, and `ClassicalFallback`. `train/`, `viz/`, `cli.py`, `app/` provide training, plotting/report, the CLI, and a Streamlit UI.

Fixed I/O contract. Inputs are always the 7 channels (`sdf, mask, x, y, uin, vin, log Re`); outputs always the 4 channels (`u, v, p,  $\nu_t$`). Fields are `(ny, nx) float32`; network tensors channel-first (`B, C, ny, nx`). Pressure is kinematic; residuals are computed on denormalised fields with $\nu_{\text{eff}} = \nu + \nu_t$. Fixing this contract is what makes the backbones interchangeable and the trust layer backbone-robust.

Backbones. `FN02d` is a faithful spectral FNO; `GeoFNO` adds a geometry-gated conditioning of the lifted features; `PhysicsTransformer` implements Transolver-style physics-slice attention (linear in grid points); `UNet/DeepONet` are baselines.

Training loss. The `CompositeLoss` sums a masked data MSE over fluid cells, a differentiable physics-residual term on the denormalised fields, and a no-slip BC term. The corrector is trained separately with the backbone frozen (subsection 3.4).

Synthetic potential-flow generator (zero-download reproducibility). `SyntheticRANS` superposes a Hess–Smith source-panel potential core (with a Kutta-enforcing bound vortex), an algebraic near-wall no-slip ramp and Gaussian wake deficit, kinematic Bernoulli pressure, and a mixing-length ν_t , with desingularised kernels and light smoothing. The fields are physically plausible and continuity-respecting but **analytic**—a smoke-test/reproducibility substrate, not solver ground truth. **No quantitative claim in this paper rests on synthetic data**; all reported numbers (section 5) are on real AirfRANS.

AirfRANS loader. `load_airfrans` reads each simulation’s point cloud, reconstructs the airfoil loop, rasterises the targets onto a structured crop, and returns (`FlowCase`, `FlowField`) pairs for all four splits.

Use of AI-assisted development tools. Substantial portions of the research software (model implementations, training/evaluation harnesses, and analysis scripts) were developed with the assistance of a generative-AI coding tool (Claude, Anthropic; Claude Opus 4.8 and Claude Fable 5 models, used through the Claude Code environment), operating under the authors’ direction. The authors specified, reviewed, and validated all code; every reported number is produced by a committed script in the public repository

Table 1: SOTA Transolver backbone (7.35M params, 80 ep, AirfRANS full), alone vs. with the DEQ correction loop. 3 seeds, mean $\pm$ std (population std, ddof=0). Lower MSE is better; ρ closer to 1 is better; resid $\leftrightarrow$ err $\rho > 0$ supports the trust signal; conformal coverage targets 0.90. Source: results/v2/v2_results.json. See subsection 5.2.

metric	mse_u	mse_v	mse_p	C_l rel. err	C_d rel. err	resid $\leftrightarrow$ err ρ
backbone	0.127 $\pm$ 0.009	0.100 $\pm$ 0.004	630 $\pm$ 42	5.6%	7.5%	0.625 $\pm$ 0.019
+ DEQ loop	0.116 $\pm$ 0.003	0.079 $\pm$ 0.005	471 $\pm$ 15	5.2%	6.7%	0.635 $\pm$ 0.021
Δ	−9%	−21%	−25%	−0.4pp	−0.8pp	+0.010

and verified against an artifact manifest of SHA-256 hashes (results/MANIFEST.json), and the full test suite accompanies the release. The scientific questions, experimental design, pre-registered hypotheses, and interpretation of results are the authors’ own.

5 Experiments

All quantitative results are on **real AirfRANS**. Unless noted, the in-distribution and out-of-distribution ablations use the pre-registered protocol (benchmarks/ablation.py, docs/EXPERIMENTS.md) over **3 seeds (0, 1, 2)**, reported as mean $\pm$ std (population std, ddof=0; sample std at $n = 3$ widens bars by $\approx 1.22\times$). Metrics follow the AirfRANS community protocol (evaluate_cases): per-channel volume MSE (lower is better, well-conditioned), surface-pressure MSE on the body, Spearman rank correlation of the force coefficients ρ_{C_l}, ρ_{C_d} (closer to 1 is better—what early-design ranking needs; throughout the tables these correlate forces integrated from the predicted vs. the ground-truth field through our grid surface-integrator—a *field-to-field self-consistency* valid for the relative arm/condition comparisons, not agreement with AirfRANS’s official force labels; see subsection 5.2), and **residual_error_spearman** (> 0 means a low residual tracks low error). The ν_t channel is near-degenerate at this scale (MSE $\approx 5 \times 10^{-8}$ for every arm) and is omitted from the tables (see section 7).

With $n = 3$ we report **per-seed sign-consistency** plus effect-size magnitude rather than p-values, which are meaningless at $n = 3$. An effect called “robust” below is sign-consistent across all 3 seeds with a large effect size and (where stated) non-overlapping per-seed distributions.

5.1 Setup

The backbone is an FNO trained on AirfRANS full (800 train / 200 test, 80 epochs). Four arms: **backbone**, **backbone (no physics loss)**, **backbone + local corrector** (feed-forward, with the acceptance test), and **backbone + DEQ corrector**. The certificate runs (subsection 5.6) use a dropout-enabled FNO (width 48, 4 layers, 20 modes, dropout 0.05) with a DEQ corrector ($\kappa = 0.9$, damping 0.5), trained 40 + 15 epochs at resolution 128. The v2 phase (subsection 5.2) re-runs the trust layer and DEQ corrector on a SOTA Transolver backbone under the same protocol; subsection 5.3–5.6 are the weak-grid-backbone runs that establish the backbone-dependence finding.

5.2 Backbone-agnostic trust + learned correction on a SOTA backbone (v2)

We re-run the trust layer and the DEQ corrector on a **SOTA Transolver backbone** (7.35M parameters, 80 epochs, AirfRANS full, 3 seeds; corrector trained 20 epochs with the backbone frozen), the strongest model in our study. The protocol and scoring are identical to subsection 5.3; the source is results/v2/v2_results.json. Table 1 reports the backbone alone and with the DEQ correction loop.

Learned correction works on a strong backbone. The DEQ corrector reduces volume field MSE on **all three seeds**: mse_u −9% (0.127 $\rightarrow$ 0.116), mse_v −21% (0.100 $\rightarrow$ 0.079), mse_p −25% (630 $\rightarrow$ 471). The per-seed reductions are sign-consistent down on all three seeds for every volume channel. **Surface-pressure** MSE is the one channel the corrector does *not* improve on this strong backbone: essentially flat

at $10794 \rightarrow 10811$ ($+0.2\%$; per-seed $-1/-1/+2\%$), which we report for completeness rather than omit. Internal force-ranking consistency is held at $\rho_{C_l} = 0.999$, $\rho_{C_d} 0.995 \rightarrow 0.996$, with relative force errors $C_l 5.6\% \rightarrow 5.2\%$, $C_d 7.5\% \rightarrow 6.7\%$. This is the corrector trained by supervised regression toward ground truth and applied without the acceptance test (subsection 3.4).

What these force correlations measure—and what they do not. Our ρ_{C_l}/ρ_{C_d} (here and in every table) correlate the force integrated from the *predicted* field against the force integrated from the *ground-truth* field through the same grid surface-integrator: a field-to-field consistency, valid for the relative arm/condition comparisons we draw from it, but *not* agreement with AirFRANS’s official OpenFOAM force labels (which our pipeline never loads). Recomputed against those official labels (`scripts/recompute_force_vs_official.py`; 200 cases, 3 seeds; `results/control/force_vs_official.json`), the deployed backbone recovers force *ranking*— $\rho_D = 0.84 \pm 0.01$ (drag), $\rho_L = 0.88$ (lift)—but not magnitude: absolute drag is biased ($\approx 11\times$ mean relative error) by the near-field integrator’s 1.5-cell surface-sampling offset, and the same integrator on *perfect* fields also yields $\rho_D = 0.84$, so the model saturates the measurement ceiling rather than the prediction being the limit.

Repairing the measurement. We then repaired what is repairable (`scripts/design_force_integrator.py`, `scripts/recompute_force_multischeme.py`; `results/control/force_vs_official_multischeme.json`). A wall-extrapolated pressure scheme cuts the ground-truth-field lift magnitude error from 16% to 4.6% (median), and a *control-volume* (far-field momentum-balance) integrator over the outer grid ring recovers official lift from ground-truth fields almost exactly ($\rho_L = 1.00$, median error $< 1\%$) and—the deployment-relevant number—from *predicted* fields at $\rho_L = 0.998 \pm 0.001$ with per-seed median magnitude errors of 3.6–3.9%: engineering-grade lift from the surrogate. Drag decomposes diagnostically: on ground-truth fields the control-volume estimator reduces the median drag error from $3.3\times$ to $0.23\times$ (the *measurement* is fixable), but on predicted fields it reaches only $\approx 2.7\times$ with a degraded ranking, so the remaining drag error is attributable to the *model’s* far-field/wake accuracy, not the integrator; drag *ranking* is still best served by the near-field scheme ($\rho_D = 0.84$). We therefore report lift as quantitatively recovered, read ρ_{C_l}/ρ_{C_d} in the tables as internal consistency, and still make no claim of accurate absolute *drag* or of a cross-method force-ranking win (published methods use a different force-integration convention). The backbone is itself a published SOTA architecture (Transolver, Wu et al. 2024), so the trust layer and corrector are demonstrated *on* a competitive surrogate—not only on the weak grid backbone.

The gain comes from the learned correction, not from feeding it the residual (W1 control). The corrector improves the field by $-9/-21/-25\%$ (gate-verified). To isolate *why*, we train the same DEQ corrector two ways under identical architecture, data, optimiser, schedule and seeds, differing only in whether the 3 residual input channels carry the real residual (WITH) or are permanently zeroed at both train and inference (NULL). Across 3 seeds, the null corrector **matches or slightly exceeds** the deployed one: the per-seed paired difference $d = \text{gain}_{\text{WITH}} - \text{gain}_{\text{NULL}}$ is negative on all 3 seeds for both `mse_u` and `mse_speed` (mean $d \approx -0.005$ on a ≈ 0.11 base, $\approx 4\%$; WITH beats NULL on 0/3 seeds; `results/control/w1_capture.json`). The residual input is therefore *not* the source of the corrector’s field-MSE improvement; the improvement is attributable to the learned correction architecture, not to residual-conditioning. This is consistent with the residual’s role elsewhere in the paper: it tells you *where* a prediction is wrong (a trust signal), not *how* to fix it. Since NULL and WITH are statistically tied (NULL marginally ahead, 0/3 seeds for WITH), the residual input is not deployed *for accuracy*; we retain it in the as-trained, residual-fed configuration only because the residual is already computed for the trust layer and the iteration’s acceptance test, so conditioning the corrector on it costs nothing extra—and we note that a residual-free corrector is an equally valid deployment. We report this as a sharpening of *attribution*, not a change to the result.

The trust signal is backbone-robust. On this SOTA backbone the bare residual-error correlation is already strong (0.625 ± 0.019 , vs ≈ 0.40 for the bare weak backbone in Table 3); the corrector barely moves it (0.635 ± 0.021), because the strong backbone’s residual is already a good detector without correction. The lift-by-corrector effect we report in subsection 5.3 is a property of the *weak* backbone; on the

Table 2: Conformal calibration of the pressure channel on the SOTA Transolver backbone ($\alpha = 0.1$, 100/100 cal/test split): MC-dropout vs. a deep ensemble ($M=5$). Both attain target coverage; only the deep ensemble yields an input-adaptive band (finite q , low ECE). MC-dropout: 3 seeds (mean $\pm$ std); deep ensemble: single ensemble training (split-to-split spread over 20 calibration/test re-draws: coverage 0.899 ± 0.009 , $q = 2.26\pm 0.05$; [subsection 5.6](#)). Sources: `results/v2/v2_results.json`, `results/uq_ensemble/uq_results.json`.

σ estimator	coverage (target 0.90)	q	ECE ($\downarrow$)	band
MC-dropout ($p=0.05$)	0.902 ± 0.008	$\approx 1.6 \times 10^9$	≈ 0.31	near-constant
deep ensemble ($M=5$)	0.915	2.35	0.074	adaptive

strong backbone the signal is strong from the start. We add a third, architecturally-distinct data point: a message-passing GNN (MeshGraphNet, param-matched at 7.35M, 3 seeds, bare backbone), where the bare case-level residual–error Spearman correlation is 0.851 ± 0.058 (`results/mgn/mgn_results.json`). Across three fundamentally different architectures—a spectral grid Geo-FNO (≈ 0.40 bare), an attention/point-cloud Transolver (0.625 ± 0.019 bare), and a message-passing GNN MeshGraphNet (0.851 ± 0.058 bare)—the correlation is consistently *positive*, though variable in strength (a 0.40–0.85 range). This is precisely what “backbone-robust” means here: the detector *works* on every architecture, not that its strength is uniform across them. (The high MeshGraphNet value partly reflects that model’s wide error spread—large, dispersed errors are easier to rank by residual.) MeshGraphNet is run only as a bare backbone (no corrector and no conformal layer), so it speaks only to the trust-signal claim. Either way the residual is a reliable detector.

Conformal coverage holds under both uncertainty estimators. Split-conformal calibration of the pressure channel at $\alpha = 0.1$ attains target coverage on this SOTA backbone with *either* uncertainty estimator: 0.902 ± 0.008 with MC-dropout (3 seeds; `results/v2/v2_results.json`) and 0.915 with a deep ensemble (single run; `results/uq_ensemble/uq_results.json`), both at the 0.90 target. Coverage was never the difficulty—**adaptivity** was. With MC-dropout this near-deterministic model has near-degenerate σ at dropout 0.05, so coverage is delivered by a **near-constant** band: the conformal quantile q is enormous ($\approx 1.5\text{--}1.7 \times 10^9$) and the reliability ECE is ≈ 0.31 .

A deep ensemble recovers an adaptive band on the SOTA model. Replacing MC-dropout with a deep ensemble of $M=5$ independently-trained Transolver members—using the per-cell standard deviation across members as σ —converts the near-constant band into a genuinely input-adaptive one while preserving coverage ([Table 2](#)). On the same 100/100 calibration/test split, the conformal quantile drops from $\approx 10^9$ to $q = 2.35$, the reliability ECE improves from ≈ 0.31 to **0.074**, and coverage holds at 0.915 (target 0.90, consistent with the $\approx \pm 0.03$ binomial spread at $n_{\text{test}} = 100$). This is a single ensemble training run, not a 3-seed average; the calibration/test *split*, however, is not the fragile part—re-drawing it 20 times leaves coverage at 0.899 ± 0.009 with $q = 2.26\pm 0.05$ on pressure (split-resampling study, [subsection 5.6](#)), the published split sitting at the upper end of that spread. The ensemble mean is also more accurate than any single member ($\rho_{C_l} = 0.9995$, $\rho_{C_d} = 0.998$, C_l/C_d relative force error 4.4%/5.7%; `results/uq_ensemble/uq_results.json`). The practical recipe follows: MC-dropout suffices when a constant-width certificate is acceptable, and a deep ensemble is what makes the band track the error on a near-deterministic SOTA surrogate.

Corrector vs. ensemble (cost, stated plainly). The ensemble mean (`mse_u/v/p` = 0.077/0.058/440) is in fact *more accurate* than the DEQ-corrected single backbone (0.116/0.079/471), so we do not claim the corrector is the best available accuracy mechanism. The two answer different questions on a cost axis: the deep ensemble is the better *accuracy* play when $5\times$ the training (and $M\times$ the inference) is affordable, while the learned corrector is the better *cheap* play—one backbone plus a small corrector—demonstrating cheap learned correction on a competitive surrogate, not a leaderboard entry. The ensemble and the corrector are complementary: the ensemble supplies the adaptive σ the conformal certificate needs, and either prediction can be wrapped by the same trust layer.

Table 3: In-distribution AirFRANS `full` ablation, 3 seeds, mean $\pm$ std (population std). Lower MSE is better; ρ closer to 1 is better; $\text{resid} \leftrightarrow \text{err } \rho > 0$ supports the trust signal. Bold marks the best arm per column among the physics-loss arms; the physics-loss-free arm (starred where best overall) is reported separately because it removes the physics objective entirely. See Figure 3.

arm	mse_u	mse_v	mse_p	surf. mse_p	ρ_{C_l}	ρ_{C_d}	resid $\leftrightarrow$ err ρ
backbone	3.479 $\pm$ 0.105	0.385$\pm$0.012	2444.8$\pm$120.7	548823 $\pm$ 27946	0.9868 $\pm$ 0.0005	0.895 $\pm$ 0.013	0.397 $\pm$ 0.003
backbone (no phys. loss)	1.995$\pm$0.042	0.323 $\pm$ 0.008*	1963.5 $\pm$ 55.5*	1123649 $\pm$ 104822	0.9920$\pm$0.0002	0.945$\pm$0.008	0.605 $\pm$ 0.016
backbone + local corr.	3.482 $\pm$ 0.058	0.398 $\pm$ 0.007	2469.4 $\pm$ 71.0	541534 $\pm$ 20453	0.9854 $\pm$ 0.0017	0.888 $\pm$ 0.012	0.389 $\pm$ 0.031
backbone + DEQ corr.	3.457 $\pm$ 0.811	0.880 $\pm$ 0.064	3832.7 $\pm$ 331.3	361681$\pm$12273	0.9856 $\pm$ 0.0017	0.923 $\pm$ 0.014	0.827$\pm$0.002

5.3 The weak grid backbone: corrector trades accuracy, lifts the trust signal

This ablation uses the **weak grid backbone**; it establishes the backbone-dependence we report as a finding, and should be read against the SOTA-backbone result in subsection 5.2, where the same corrector *does* reduce field MSE.

On the weak backbone the corrector trades accuracy. The DEQ corrector is **flat** on volume `mse_u` (3.457 vs backbone 3.479; the per-seed deltas are sign-inconsistent, 2/3, so this is noise, not an improvement) and **worse** on `mse_v` (0.385 $\rightarrow$ 0.880, +129%, 3/3, non-overlapping distributions) and `mse_p` (2445 $\rightarrow$ 3833, +57%, 3/3). The feed-forward `local` corrector helps on **nothing**: it is at-or-worse than the backbone on every volume MSE channel and 3/3 worse on ρ_{C_d} . On this weak backbone the residual-conditioned correctors do not improve volume accuracy; the DEQ arm buys surface-pressure MSE (−34%, 3/3) and a slightly better drag ranking (ρ_{C_d} 0.895 $\rightarrow$ 0.923, 3/3) at a measured cost to volume velocity and pressure. We do **not** claim the corrector improves accuracy in aggregate *on this backbone*—and this is precisely the contrast that makes correction quality backbone-dependent.

The detector succeeds. The same DEQ arm roughly **doubles** the residual-error Spearman correlation, 0.397 $\rightarrow$ 0.827 (3/3, std $\approx$ 0.002, Cohen’s $d \approx$ 143). A low residual tracks low field error far more reliably with the corrector engaged. This is the detector half of the thesis, and it is the largest, cleanest effect in the table.

A control that sharpens the thesis. Removing the physics loss entirely (`backbone (no physics loss)`) **improves** `mse_u`, `mse_v`, `mse_p`, ρ_{C_l} , ρ_{C_d} , and the residual-error correlation (all 3/3), at the cost of **doubling** surface-pressure MSE (549k $\rightarrow$ 1124k, 3/3). The physics-loss-free backbone is in fact the **best force-ranking model in the table** (ρ_{C_d} 0.945, ρ_{C_l} 0.992)—better than any corrected model, with no correction loop at all. We therefore explicitly do **not** claim the loop delivers best-in-class force ranking. The physics objective (in the loss or the corrector) buys surface-pressure fidelity and a strong trust signal, not volume accuracy or force ranking *on this weak backbone*—the trust-signal role succeeds while the correction role does not, which is exactly the backbone-dependence the SOTA result in subsection 5.2 resolves.

Retraction (retained from prior drafts). An earlier single-seed run reported ρ_{C_l} rising 0.924 $\rightarrow$ 0.958 and surface MSE −25% under the corrector. These do **not** replicate: across 3 seeds the backbone already sits at ρ_{C_l} = 0.987 and the DEQ corrector is 0.986 (flat-to-slightly-worse, sign-inconsistent). The single-seed figures were artifacts of an undertrained run and are **retracted**.

5.4 Residuals detect under distribution shift

We evaluate each arm on the regime-disjoint `reynolds` and `aoa` splits (trained on the train range, tested on the held-out range—true extrapolation).

The trust signal is regime-invariant—the single strongest result. On the `aoa` split the bare backbone’s residual-error correlation **collapses to** 0.314, while the DEQ-corrected model holds it at 0.746 (0.314 $\rightarrow$ 0.746, 3/3, Cohen’s $d \approx$ 8.9, non-overlapping: every corrected seed $\geq$ 0.72, every backbone seed $\leq$ 0.38). On `reynolds` both are high ($\approx$ 0.74–0.75). The corrected model thus keeps a regime-invariant

Table 4: Out-of-distribution AirfRANS ablation (regime-disjoint train→test), 3 seeds, mean $\pm$ std. The full four-arm table is in `results/full_research/ood/ablation_ood.md`. See Figure 4.

task	arm	mse_u	mse_v	mse_p	surf. mse_p	ρ_{C_l}	ρ_{C_d}	resid↔err ρ
reynolds	backbone	16.218 $\pm$ 0.746	1.187 $\pm$ 0.149	7220.7 $\pm$ 612.2	964641 $\pm$ 61419	0.950 $\pm$ 0.008	0.893 $\pm$ 0.009	0.748 $\pm$ 0.077
reynolds	+ DEQ	5.300 $\pm$ 1.544	1.208 $\pm$ 0.255	6469.5 $\pm$ 438.0	652281 $\pm$ 45390	0.943 $\pm$ 0.008	0.915 $\pm$ 0.009	0.742 $\pm$ 0.041
aoa	backbone	4.312 $\pm$ 0.077	1.310 $\pm$ 0.039	6042.8 $\pm$ 242.5	2000592 $\pm$ 55092	0.963 $\pm$ 0.003	0.926 $\pm$ 0.002	0.314 $\pm$ 0.064
aoa	+ DEQ	3.538 $\pm$ 0.681	1.407 $\pm$ 0.089	4620.6 $\pm$ 170.2	1042711 $\pm$ 78298	0.966 $\pm$ 0.006	0.905 $\pm$ 0.013	0.746 $\pm$ 0.027

≈ 0.74 trust signal across both shifts, precisely where the one-shot model loses it. The residual is a reliable detector exactly where detection matters most.

Scoping the rest, honestly (protected negatives). The OOD ablation also shows the DEQ corrector reducing the in-distribution→OOD *gap* on volume velocity, volume pressure, and surface pressure (e.g. reynolds mse_u 16.2 $\rightarrow$ 5.3, aoa surf. mse_p 2.00M $\rightarrow$ 1.04M, both 3/3). We report this as an observation about the loop’s behaviour, **not** as a competitiveness or generalisation guarantee, and we explicitly preserve two negatives. (i) **mse_v is an attenuated-regression artifact, not a gain:** the DEQ corrector’s *absolute* OOD mse_v is worse than the backbone in every regime (0.880/1.208/1.407 vs 0.385/1.187/1.310); its in-dist→OOD gap only looks smaller because its in-distribution mse_v is already inflated. (ii) **Drag ranking is task-dependent:** ρ_{C_d} improves on reynolds (0.893 $\rightarrow$ 0.915, 3/3) but **regresses on aoa** (0.926 $\rightarrow$ 0.905, worse on all 3 seeds). We do not claim the loop uniformly preserves force ranking under shift. Methodological caveat: the in-distribution reference is the full split (a different training range), so the gap mixes a train-set change with regime shift, and the aoa mse_u effect, while 3/3 directional, is seed-0-dominated.

A second dataset: the trust signal generalizes beyond AirfRANS. The strongest test of generality is a different dataset entirely. We evaluate the residual trust signal on DeepCFD (Ribeiro et al., 2020)—2-D laminar incompressible flow over bluff bodies (circles, squares, rhombi): a different geometry family *and* a different regime (laminar, versus AirfRANS’s turbulent RANS). We train an FNO backbone (3 seeds) on an out-of-distribution split that holds out the largest bodies—forcing a range of per-case errors—and measure the same per-case residual–error Spearman correlation (interior continuity+momentum residual; `scripts/run_deepcfd.py`, `results/deepcfd/deepcfd_results.json`). The trust signal holds (Figure 9): **residual_error_spearman** = 0.770 $\pm$ 0.119 (3 seeds), comparable to—indeed above—the AirfRANS Transolver value (0.625). The error spread is genuine (per-case rel- L_2 median 0.063, max 0.136), so the correlation is not degenerate; per-seed case-level bootstrap 95% CIs (10^4 resamples) are [0.65, 0.77], [0.58, 0.72] and [0.91, 0.95], all far from zero (`results/control/bootstrap_spearman_ci.json`). A clean mechanistic reason it can be *sharper* than the turbulent case: laminar flow has $\nu_t = 0$, so the residual is the *exact* Navier–Stokes residual with no turbulence-closure ambiguity—a cleaner error indicator. This is direct evidence that the residual-as-trust-signal property is not specific to AirfRANS, its geometry family, or its turbulence model. (We report only the trust signal here; DeepCFD’s force coefficients carry the same self-integrator caveat as subsection 5.2 and are not headlined.)

Beyond regime shift: a geometry the model never saw (qualitative). The splits above shift the flow regime *within* the airfoil family; we close with a qualitative probe of cross-*geometry* extrapolation. We run the airfoil-trained Transolver on a *cylinder*—a bluff body absent from AirfRANS—and ask not whether the prediction is good (it is not) but whether the residual—a data-free quantity evaluated against the governing PDE, needing no reference field (unavailable for a cylinder at these Re)—registers the geometry as out-of-distribution. It does, in the residual *magnitude*: the predicted cylinder field is uniformly less PDE-consistent than in-distribution airfoils with good predictions ($\approx 0.6\%$ rel- L_2). Normalising each case’s residual by its freestream U^2 (the momentum-residual scale), the dimensionless far-field residual is $\approx 7\times$ higher on the cylinder (median $\approx 30\times$; Figure 10, `results/figures/cylinder_control_airfoil_ratio.json`). This gap holds in the *far field*, away from any wall, so it is a genuine geometry-OOD effect—neither a near-wall discretisation artifact nor a velocity-scale effect (it survives the U^2 normalisation). **Honest scope, with a controlled negative.** This is a qualitative illustration on one geometry family, and we report a control that *refutes* the tempting spatial reading: the residual’s bright near-body *ring* is a generic no-slip-wall feature,

Table 5: Iteration sensitivity sweep: `results/sensitivity/iters.csv`. Single sweep (one checkpoint, not seeded). See Figure 6.

n_iters	mse_u	surf. mse_p	residual_norm	resid $\leftrightarrow$ err ρ
0	3.924	541204	0.113	0.423
1	2.460	428560	0.336	0.644
3	2.287	336718	0.542	0.647
5	2.439	308381	0.594	0.675
10	2.570	300298	0.618	0.703
15	2.575	300664	0.620	0.710

not an OOD localiser—in-distribution airfoils with good predictions show the *same* near-body-vs-far-field concentration ratio (≈ 23) as the cylinder (≈ 18), so spatial near-wall concentration cannot be read as “where the prediction failed.” The out-of-distribution signal is therefore the field-wide residual *magnitude*, not its spatial ratio; and the residual remains a local-violation detector, not a global-correctness oracle. It complements, not replaces, the quantitative regime-shift results above.

5.5 Residual-as-objective fails: the residual and the error move apart

The most direct evidence that the residual is a poor *objective* comes from sweeping the number of correction iterations on a single trained checkpoint (Table 5).

As the loop iterates, the **PDE residual norm rises monotonically** ($0.11 \rightarrow 0.62$) while the **field error falls** (`mse_u` $3.92 \rightarrow 2.29$ by iter 3)—the two objectives move in *opposite* directions. This is the clean statement of detector $\neq$ fixer: the very quantity one would minimise to “fix” the field grows while the field gets better. (Note this does not contradict the DEQ contraction of subsection 5.6: the contraction is in the correction variable δ , which converges; the PDE residual of the *output field* is a different quantity, and it is the one that rises.)

The acceptance test accepts almost nothing. On the feed-forward corrector the backtracking acceptance test—which admits only residual-reducing steps—is the honest “certified” version of the loop. Because the test is satisfiable by the identity (zero step), a corrector that cannot reduce the *residual* is simply rejected, and the feed-forward corrector—flat-to-worse on every accuracy metric in Table 3—has essentially no residual-reducing step to offer, so the certified loop makes almost no accepted progress. The corrector that *does* help on the design metrics (DEQ) is precisely the one applied **without** the acceptance test (subsection 3.4). The “certified self-correction” guarantee is therefore real but vacuous as an accuracy mechanism: it correctly refuses to make the residual worse, and in doing so does almost nothing. (This structural argument rests on the acceptance test’s definition plus the Table 3 feed-forward result; we do not have a committed artifact that quantifies the accepted-step count, and flag it as the thinnest leg of the residual-as-objective negative.) We additionally note that for the DEQ corrector the trust-gating and acceptance-test toggles are structural no-ops—the DEQ branch bypasses both—so the only live correction knob is the applied-delta step size (`results/sensitivity/toggles.json`).

5.6 The conformal trust layer: coverage and contraction

Contraction (H5). Iterating the learned DEQ operator $\delta_{k+1} = T_\theta(\delta_k)$ from a random initialisation on 24 real AirFRANS cases, the measured per-step ratio $\|\delta_{k+1} - \delta^*\|/\|\delta_k - \delta^*\|$ (in the geometric regime, before the $\sim 10^{-7}$ solve floor) has **median 0.78** and maximum 0.86—strictly below the design bound $\kappa = 0.9$ and the falsification threshold 1—and reaches a relative distance of 10^{-5} to the fixed point in a median of 37 steps (`results/certificates/h5_contraction.json`). The corrector contracts as designed; this is a property of the *operator*, separate from whether minimising the output’s PDE residual helps the field (it does not, subsection 5.5).

In-distribution coverage (H4). With per-cell MC-dropout σ (16 passes) and a 100/100 calibration/test split of the AirfRANS test set, split-conformal calibration at $\alpha = 0.1$ attains per-channel coverage of **0.911** (u), **0.928** (v), **0.942** (p)—all in the $[0.85, 0.95]$ band and conservatively above the 0.90 target, as the $\geq 1 - \alpha$ guarantee requires. The conformal multipliers ($q = 0.77, 1.30, 1.75$) both tighten an over-dispersed and widen an under-dispersed raw σ , so the calibration is doing real work. Coverage tracks $1 - \alpha$ across a sweep of α (Figure 8). The reliability diagram is exact at the target level but over-covers at lower nominal levels (ECE $u/v/p = 0.064/0.093/0.130$), the signature of a heavy-tailed $|\text{error}|/\sigma$ ratio; we therefore claim coverage at the chosen α , **not** full distributional calibration (`results/certificates/h4_coverage.json`).

The certificate is calibrated on the deployed, corrected field—not the raw backbone. NeuroForge ships the DEQ-corrected field, so the coverage guarantee must hold for *that* output; split-conformal calibration, however, applies to whatever predictor it scores, and the raw-backbone certificate does **not** transfer unchanged through the corrector. We verify this on the dropout-FNO backbone whose DEQ corrector materially changes the field (mean $|\text{corrected} - \text{raw}| \approx 5$), through the `same_cases_to_error_sigma` path as H4 but with a corrected predict function (`scripts/verify_conformal_corrected_field.py`; 15/15 calibration/test). The clean isolation is a *within-run frozen- q contrast*—same fitted q , same input-conditional σ , same cases, only the raw vs. corrected error map—and it shows the direction of the coverage change tracking the direction of the error change: where the corrector *enlarges* the error the raw band drops below the 0.90 target (v 0.896 $\rightarrow$ 0.819, p 0.864 $\rightarrow$ 0.777), and where it *shrinks* the error the raw band over-covers and stays conservative (u 0.856 $\rightarrow$ 0.888). This is the coverage analogue of detector $\neq$ fixer (subsection 5.5): the DEQ operator contracts in its *own* variable δ (H5, median ratio 0.78), but the Banach bound governs the correction, not $|\text{error}|/\sigma$, so it cannot imply coverage survival—and on v/p residual contraction coincides with error *growth*, which is exactly what breaks the raw band. Re-fitting q on the corrected calibration field—at no change to σ , which is input-conditional and identical for both fields—restores coverage toward target where it had collapsed (v 0.819 $\rightarrow$ 0.894, p 0.777 $\rightarrow$ 0.834) and sharpens it where it was loose (u 0.888 $\rightarrow$ 0.874). We therefore calibrate the shipped certificate on the corrected output; split-conformal validity requires only exchangeable corrected-field scores, not that σ be the corrected field’s own uncertainty. These estimates are directional at $n = 15$ (per-case coverage std ≈ 0.15 –0.23, so $\approx \pm 0.05$ standard error), but the same pattern—raw- q under-covers v/p on the corrected field, refitting restores it—reproduces on an independent 20/20 split (`results/certificates/probe_conformal_after_deq.json`); the residual p under-coverage is the small-sample heavy-tail effect already noted at $n = 100$. On the deployed SOTA Transolver backbone we now verify this *directly* rather than by extrapolation: through the same conformal path, with σ the frozen five-member deep-ensemble std and the scored field replaced by the DEQ-corrected output, at the production 100/100 split and across all three corrector seeds (`scripts/run_w2_conformal_corrected.py`, `results/uq_ensemble/w2_conformal_corrected.json`; a faithfulness gate reproduces the published backbone p -certificate— $q=2.35$, coverage 0.915—exactly). Re-fitting q on the corrected calibration field gives a valid certificate (coverage $u/v/p = 0.88/0.91/0.91$ vs the 0.90 target), and because the corrected field’s error is lower at frozen σ , that accuracy propagates into the certificate as proportionally tighter intervals at the same coverage (q : v 2.58 $\rightarrow$ 2.34, p 2.35 $\rightarrow$ 2.10 on every seed). Applying the *backbone’s* own q to the corrected field—the direct “remains-conservative” test—over-covers on v and p (≈ 0.93), as the field-error reduction (subsection 5.2) predicts, while u holds at ≈ 0.88 , just below target: this tracks the backbone’s own heavy-tailed u band (0.881) and is inherited from it, not introduced by the corrector. We therefore ship the certificate refit on the corrected output; recalibration matters most for correctors that trade field error for residual reduction (the dropout-FNO above), and is benign-to-beneficial for the deployed Transolver, whose bands it sharpens.

Split-resampling robustness (20 splits). Because the certificates above are computed on one 100/100 calibration/test draw, we re-draw the split 20 times (`scripts/run_multisplit_conformal.py`, `results/uq_ensemble/multisplit_conformal.json`; same code path, frozen ensemble σ , q refit per split). Coverage is tightly centred on the 0.90 target in *every* arm and channel (0.895–0.902, split-to-split std ≤ 0.014); the published seed-0 pressure coverage of 0.9155 sits at the upper end of its distribution (0.899 ± 0.009). The corrected-field sharpening is robust, not a split artifact: on pressure, q drops from 2.26 ± 0.05 (backbone) to 2.02–2.07 for all three corrector seeds, and on v from 2.55 ± 0.05 to 2.28–2.31; on

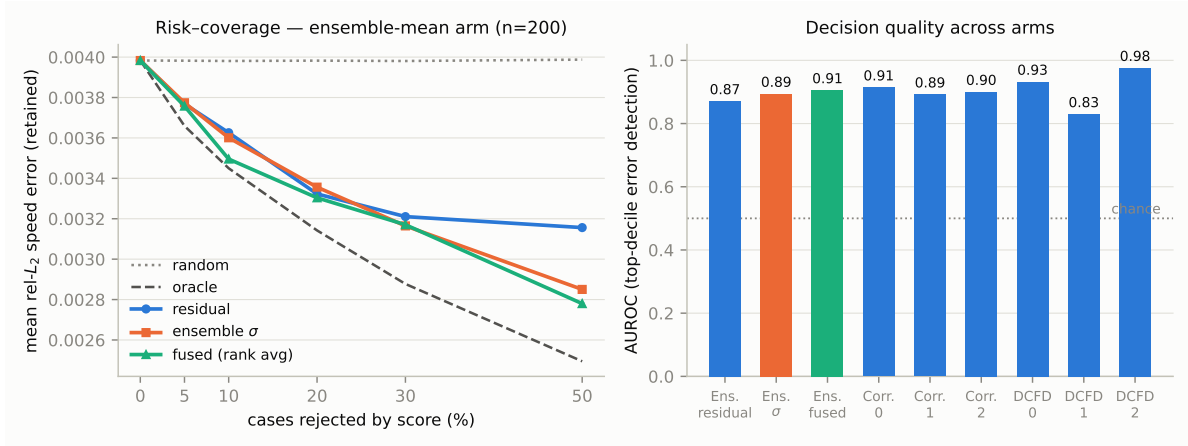


Figure 2: **Selective prediction with the trust signal.** *Left:* risk-coverage curves on the AirfRANS test set (ensemble-mean arm): mean retained relative- L_2 error after rejecting the least-trusted fraction of cases by physics residual, ensemble σ , or their rank fusion, against the oracle (reject by true error) and random rejection. *Right:* AUROC for detecting worst-decile-error cases across arms (AirfRANS ensemble-mean, three corrected seeds, three DeepCFD OOD seeds); chance = 0.5.

the heavy-tailed u channel the effect is mixed (one seed sharpens, two widen slightly), consistent with the u -neutral verdict above.

Out-of-distribution coverage (empirical, not a guarantee). Calibrating q on the in-distribution full split and evaluating on the OOD splits, coverage stays in the target band: u 0.90/0.91/0.91, v 0.91/0.94/0.92, p 0.92/0.94/0.90 (full / reynolds / aoa) vs the 0.90 target (**results/sensitivity/ood_coverage.json**, Figure 7). **This is an empirical observation, not a guarantee.** The conformal guarantee assumes exchangeable calibration and test draws, which fails under shift; coverage nonetheless holds because the MC-dropout σ inflates appropriately on shifted inputs, so the same q still brackets the (larger) errors. We report this as a desirable empirical property and explicitly **do not** assert distribution-free coverage out-of-distribution.

From correlation to decisions: selective prediction. A rank correlation of 0.6 can sound modest, so we also measure what the trust signal is *for*: deciding which predictions to reject (**scripts/run_selective_prediction.py**, **results/selective/selective_prediction.json**; Figure 2). On the 200 AirfRANS test cases (ensemble-mean arm), the residual detects worst-decile-error cases with AUROC 0.871; the ensemble σ alone gives 0.894; and the rank-fused residual+ σ score reaches **AUROC 0.905** (AUPRC 0.681 against a 0.10 prevalence baseline). The same residual score transfers: AUROC 0.89–0.91 on the three DEQ-corrected arms and 0.83–0.98 on the three DeepCFD OOD seeds. In risk-coverage terms, rejecting the 10% least-trusted cases by the fused score reduces the retained mean relative- L_2 error from 0.0040 to 0.0035—recovering $\approx 91\%$ of the error reduction an *oracle* that rejects by true error could achieve at that budget (the residual alone recovers $\approx 67\%$; random rejection recovers none). Case-level bootstrap 95% CIs (10^4 resamples; **results/control/bootstrap_spearman_ci.json**) put the ensemble-arm residual correlation at 0.610 [0.506, 0.698] and the fused score at 0.703 [0.615, 0.773]; every arm’s CI excludes zero by a wide margin. The trust signal is thus not merely correlated with error; it supports near-oracle triage at deployment-relevant rejection budgets.

The trust layer depends only on a model’s predictions and σ , so it is backbone-robust; subsection 5.2 demonstrates it on a SOTA Transolver backbone, where coverage holds at 0.902 ± 0.008 (target 0.90) and a deep-ensemble σ makes the band input-adaptive on that near-deterministic model ($q = 2.35$, ECE 0.074; Table 2).

Table 6: Matched-budget baseline on AirfRANS full ($n_{\text{train}} = 800$, 80 epochs, identical rasterisation and `evaluate_cases` scoring), 3 seeds. See Figure 5.

model	n_params	mse_u	mse_v	mse_p	surf. mse_p	ρ_{C_l}	ρ_{C_d}
Transolver (baseline)	7.35M	0.120 $\pm$ 0.005	0.088 $\pm$ 0.013	628.5 $\pm$ 29	9110 $\pm$ 500	0.9992 $\pm$ 0.0002	0.9963 $\pm$ 0.0012
our backbone (Table 3)	—	3.479	0.385	2444.8	548823	0.987	0.895
our backbone + DEQ	—	3.457	0.880	3832.7	361681	0.986	0.923

Table 7: Published AirfRANS full-task baselines from Bonnet et al. (2022), as field context only. Volume MSE on standardized fields ($\times 10^{-2}$); surface $\bar{p}$ ($\times 10^{-1}$); force relative errors (%); ρ = Spearman rank correlation vs. *official* field labels. NeuroForge is *not* a comparable row (physical-unit MSE, self-integrated forces; subsection 5.2). The near-zero/negative ρ_{C_d} is the benchmark’s known drag-ranking difficulty.

method	$\bar{u}_x$	$\bar{p}$	surf. $\bar{p}$	C_L err	C_D err	ρ_{C_l} / ρ_{C_d}
MLP	0.95	0.74	1.13	0.77	4.29	0.913 / -0.117
GraphSAGE	0.83	0.66	0.66	0.52	4.05	0.965 / -0.303
PointNet	3.50	1.15	0.93	0.74	14.64	0.938 / -0.022
Graph U-Net	1.52	0.66	0.39	0.49	10.39	0.967 / -0.138

5.7 Honest baseline: the grid backbone trails SOTA

Transolver, a SOTA point-cloud physics-attention transformer, trained on the same data with the same budget and scored identically, is far ahead of our grid backbone: relative to the bare backbone, roughly $29\times$ **on mse_u**, $4\times$ **on mse_v** ($\approx 10\times$ against the volume-regressed DEQ arm), $4\times$ **on mse_p**, and $\sim 60\times$ **on surface pressure** ($\approx 40\times$ against the DEQ arm, whose surface MSE is lower), with near-perfect force ranking. Our grid backbone is not a competitive surrogate; we make **no competitiveness claim** for it. The decisive next step follows in subsection 5.2: we put the trust layer and the DEQ corrector *on* this Transolver backbone, where the residual is a calibrated detector ($\rho = 0.625$) and the corrector reduces field MSE by 9–25%. The large grid-backbone gap is itself informative: even a much weaker backbone yields a residual that is a calibrated, shift-robust detector of its own errors.

Note on the two Transolver runs. The Transolver numbers here (Table 6: mse_u 0.120 $\pm$ 0.005, mse_v 0.088 $\pm$ 0.013, mse_p 628.5 $\pm$ 29) and the v2 backbone-alone numbers (Table 1: 0.127 $\pm$ 0.009, 0.100 $\pm$ 0.004, 630 $\pm$ 42) are the *same Transolver architecture at the same budget* (7.35M params, 80 epochs), trained in two separate runs; the differences are within the reported per-seed spread. The v2 run is the one we use as the NeuroForge backbone in subsection 5.2.

Positioning against the published AirfRANS field. Our trust layer and corrector run on a Transolver backbone—a published SOTA architecture (Wu et al., 2024)—so competitiveness comes from that architectural choice, not from re-benchmarking our own metric values. We deliberately do *not* place a NeuroForge row in a cross-method comparison: AirfRANS results are reported in mutually-incompatible conventions (the AirfRANS-paper baselines use standardized-field MSE and force coefficients against official OpenFOAM labels; Transolver-lineage papers report relative- L_2 ; our pipeline uses physical-unit MSE and self-integrated forces, subsection 5.2), so any shared cell would mislead. Table 7 reproduces the canonical AirfRANS-paper baselines as field context. Note the benchmark’s signature difficulty—drag ranking ρ_{C_d} is near zero or negative for every baseline, the headline finding of Bonnet et al. (2022)—which is also why we read our own ρ_{C_d} as internal self-consistency rather than a solved drag-ranking claim.

5.8 Figures

5.9 Reading against the pre-registered hypotheses

- **H1 (the corrector improves accuracy).** *Backbone-dependent.* On the **weak grid backbone** it is *not supported as pre-registered*: the feed-forward corrector helps on nothing and the DEQ corrector

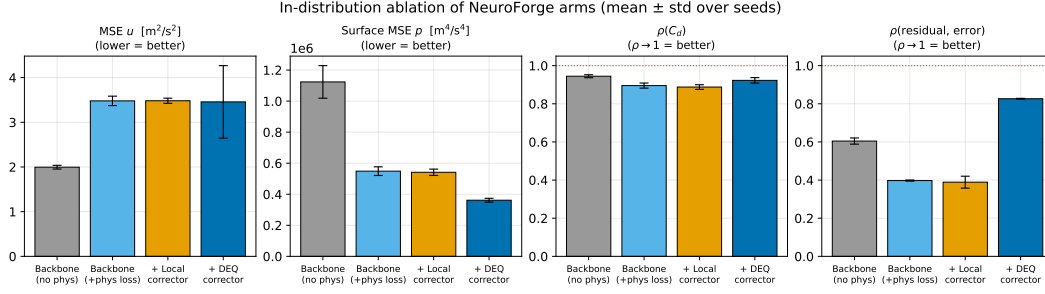


Figure 3: In-distribution ablation (Table 3): the DEQ corrector lifts the trust signal ($0.40 \rightarrow 0.83$) and surface fidelity while regressing volume $\text{mse}_v/\text{mse}_p$.

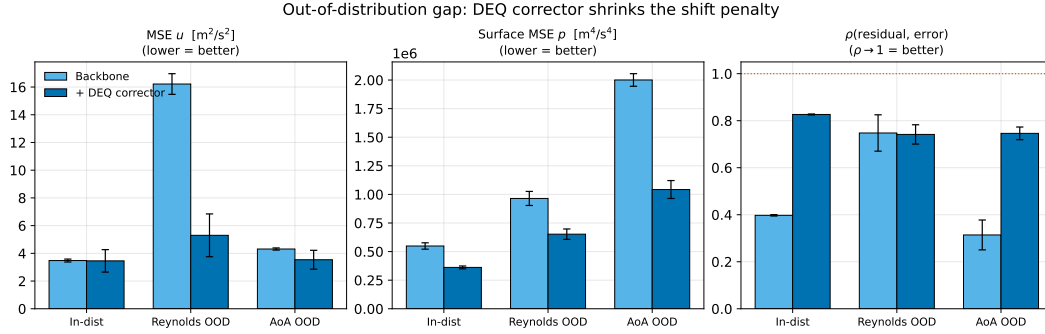


Figure 4: OOD gap and the regime-invariant trust signal ($\text{aoa } 0.31 \rightarrow 0.75$); the bare backbone’s signal collapses, the corrected model’s holds.

regresses volume $\text{mse}_v/\text{mse}_p$ (3/3) while flat on mse_u . On the **SOTA Transolver backbone** it *is supported*: the DEQ corrector reduces all three volume MSE channels on all 3 seeds (mse_u -9% , mse_v -21% , mse_p -25% ; subsection 5.2, Table 1) with force ranking held. Correction quality is therefore backbone-dependent—a finding, not a hidden negative—and the supervised corrector works where the backbone is strong.

- **H2 (the residual is a valid trust signal).** *Supported, robustly, in- and out-of-distribution.* $\text{resid} \leftrightarrow \text{err } \rho > 0$ for every arm/seed/split; DEQ lifts it to 0.83 in-dist and makes it regime-invariant (≈ 0.74) OOD, rescuing the aoa collapse $0.31 \rightarrow 0.75$.
- **H3 (DEQ $\geq$ feed-forward corrector).** *Holds on the design axis only.* DEQ beats the feed-forward corrector on surface MSE, ρ_{C_d} , and the trust signal (3/3), but is worse on volume $\text{mse}_v/\text{mse}_p$; the two trade off.
- **H4 (conformal coverage).** *Supported in-distribution* (0.91/0.93/0.94 at $\alpha = 0.1$; SOTA backbone 0.902, deep-ensemble band adaptive at ECE 0.074, Table 2); *empirical only OOD* (subsection 5.6).
- **H5 (DEQ contraction).** *Supported* (measured factor $0.78 < \kappa = 0.9 < 1$).

Every number comes from the committed, reproducible harness (benchmarks/ablation.py, scripts/run_certificates.py, results/sensitivity/).

6 The residual floor: a self-consistent monitor is a good detector but a bad fixer

The iteration sweep of subsection 5.5 (Table 5) shows a counter-intuitive fact: driving the monitored physics residual *down* drives the field error *up*. We now explain this exactly. The mechanism is not ill-conditioning

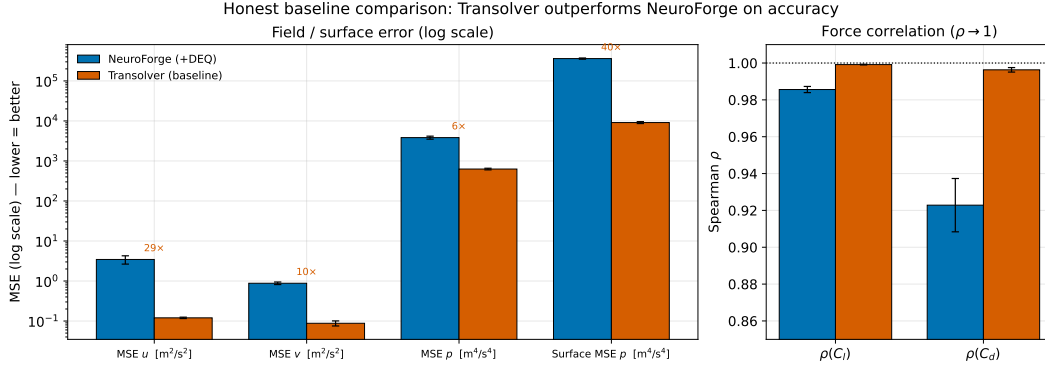


Figure 5: Honest matched-budget baseline; the grid backbone trails Transolver ~ 4 – $60\times$ across channels (no competitiveness claim).

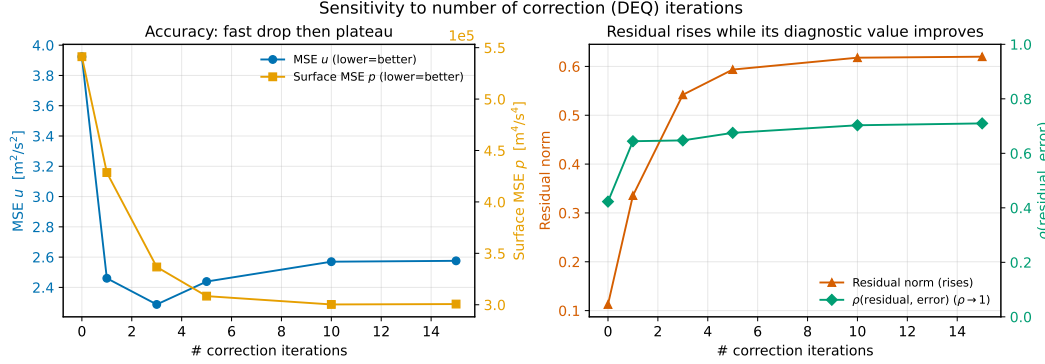


Figure 6: The money figure for the headline: PDE residual *rises* while field error *falls* across correction iterations (detector $\neq$ fixer).

per se; it is that the *monitored* residual is self-consistent at a field that is not the truth—so its minimiser sits away from u^* by an irreducible amount. The same decomposition that forces this also explains why the residual is nonetheless an excellent error *detector*: one equation, both verdicts.

6.1 Setup: the monitored operator

Let R_h be the residual operator the framework actually monitors and (in the objective/gate role) minimises: the *discrete* steady incompressible RANS **continuity and momentum** residual on the (n_y, n_x) grid, evaluated on physical fields with $\nu_{\text{eff}} = \nu + \nu_t$. By construction (`Diagnostics.residual_norm`; [section 3](#)) it **omits the no-slip wall term**. Write the objective $J(u) = \frac{1}{2} \|R_h(u)\|^2$. Let u^* be the discrete ground-truth field, define the *residual floor* $r^* := R_h(u^*)$, and for any prediction $\hat{u}$ write the error $e := \hat{u} - u^*$ with the first-order expansion

$$R_h(\hat{u}) = r^* + L e + o(\|e\|), \quad L := DR_h(u^*). \quad (7)$$

Theorem 1 (Residual floor: misplaced minimum, quantified floor, and local divergence). *Assume (H1) R_h is the monitored discrete operator above (continuity+momentum, no-slip omitted); and (H2) the floor is nonzero, $r^* := R_h(u^*) \neq 0$. Write $L := DR_h(u^*)$ and let $\sigma_{\min}$ denote its smallest nonzero singular value (no injectivity is assumed; L has a non-trivial kernel—see the remark below). Then:*

- (i) (Misplaced minimum, exact.) *The uniform-freestream field u_∞ satisfies $R_h(u_\infty) = 0$ exactly, so $\min_u J = 0$; whereas $J(u^*) = \frac{1}{2} \|r^*\|^2 > 0$. Hence u^* is **not** a minimiser of J .*

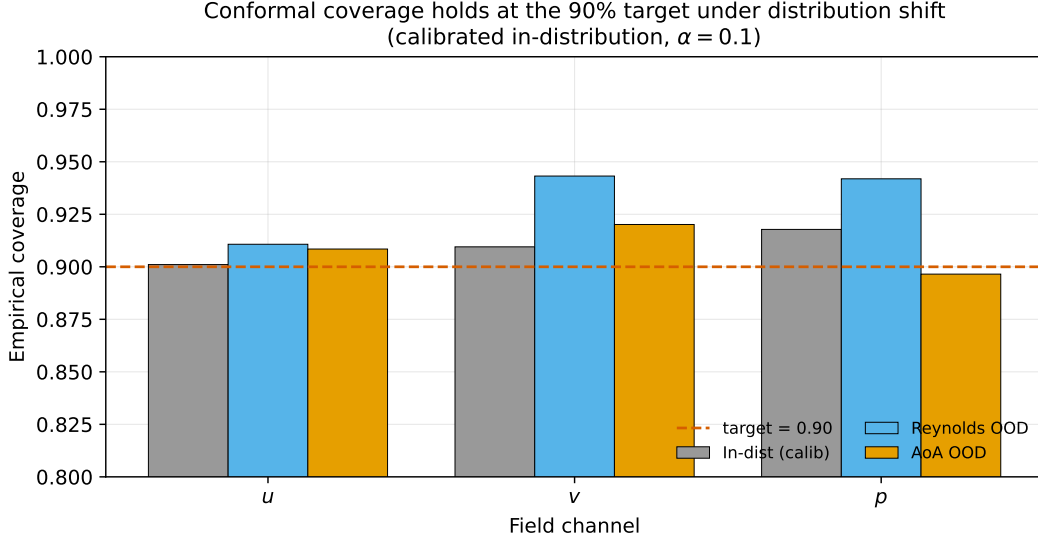


Figure 7: Conformal coverage stays in-band on OOD splits (empirical, via σ inflation).

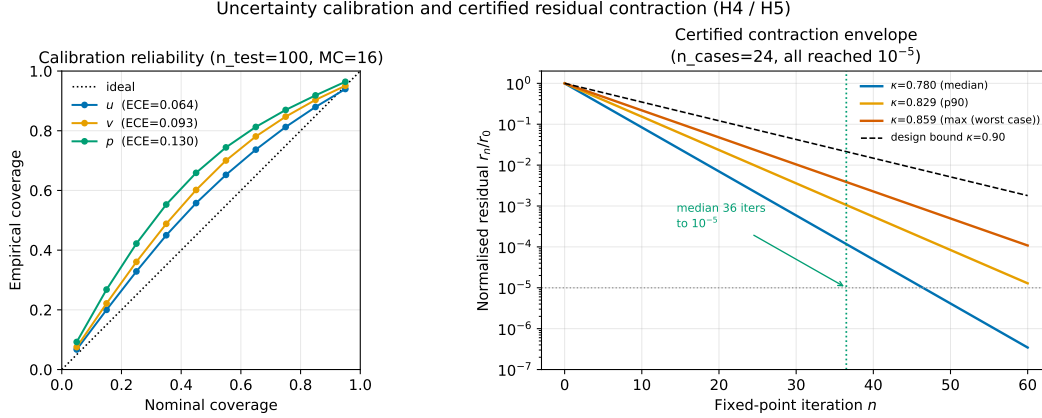


Figure 8: In-distribution reliability / coverage-vs- α and the measured DEQ contraction ($0.78 < 1$).

- (ii) (Misplaced stationarity.) $\nabla J(u^*) = L^\top r^*$, so u^* is a stationary point of J iff $L^\top r^* = 0$; if $r^* \notin \ker L^\top$, gradient flow on J leaves u^* .
- (iii) (Quantified floor.) The minimum-norm minimiser of the linearised objective $\frac{1}{2}\|r^* + Le\|^2$ is $e_\infty = -L^+ r^*$ (L^+ the Moore–Penrose pseudoinverse), leaving irreducible field error $\|e_\infty\| = \|L^+ r^*\| \leq \|r^*\|/\sigma_{\min}$, which grows as $\sigma_{\min} \rightarrow 0$ (ill-conditioning).
- (iv) (Local divergence direction.) Under residual-gradient flow $\dot{e} = -\nabla J(\hat{u})$, the field error obeys $\frac{d}{dt} \frac{1}{2}\|e\|^2|_0 = -\|Le\|^2 - (Le) \cdot r^*$. There is an open cone of error directions—those with $(Le) \cdot r^* < -\|Le\|^2$ —on which a residual-reducing step strictly **increases** the field error.

Proof. (i) For a spatially constant field every finite-difference derivative vanishes, so the discrete continuity and momentum residuals are identically zero: $R_h(u_\infty) = 0$, giving $J(u_\infty) = 0 = \min J$ (as $J \geq 0$). By (H2), $J(u^*) = \frac{1}{2}\|r^*\|^2 > 0$. (ii) $\nabla J(u) = DR_h(u)^\top R_h(u)$; evaluate at u^* and use (7). (iii) $\frac{1}{2}\|r^* + Le\|^2$ is convex in e ; its minimum-norm minimiser is the least-squares solution $e_\infty = -L^+ r^*$, which requires no injectivity (it is the Moore–Penrose pseudoinverse of the normal equations $L^\top Le = -L^\top r^*$), and $\|L^+\|_2 = 1/\sigma_{\min}$ with $\sigma_{\min}$ the smallest nonzero singular value of L . (iv) To first order $\dot{e}|_0 = -L^\top(r^* + Le)$, so $\frac{d}{dt} \frac{1}{2}\|e\|^2|_0 =$

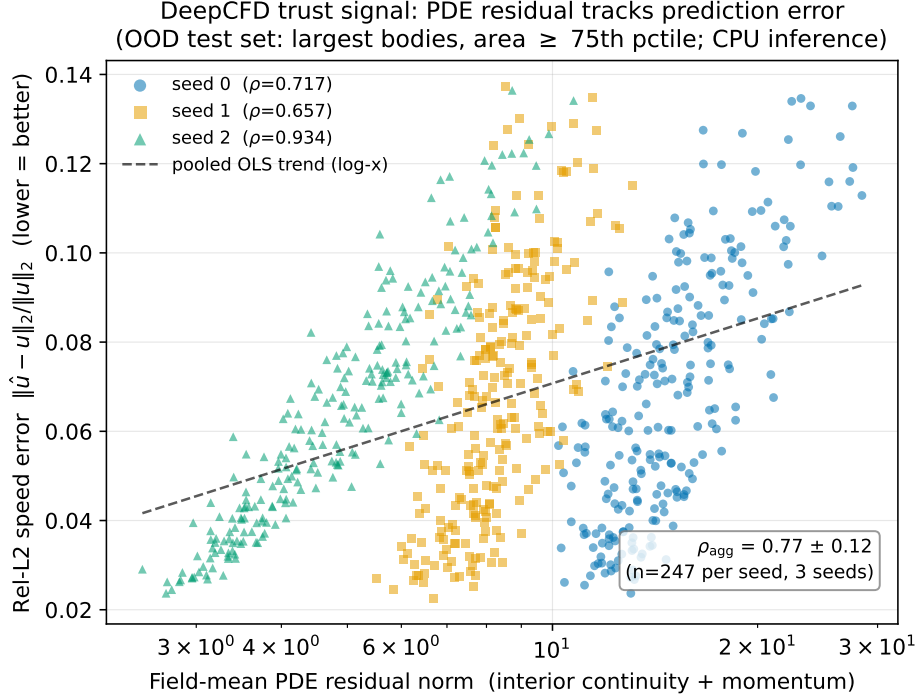


Figure 9: The trust signal generalizes to a second dataset (DeepCFD, 2-D laminar flow over bluff bodies; OOD held-out-largest-body split, 3 seeds). Per-case field-mean PDE residual norm (interior continuity+momentum) vs. rel- L_2 speed error: a low residual tracks a low error, as on AirfRANS. Per-case Spearman $\rho = 0.77 \pm 0.12$ (per seed 0.72/0.66/0.93; the trend holds *within* each seed—the colour-coded clusters span different residual ranges by initialisation). The residual is a reference-free error proxy on a geometry family and flow regime absent from AirfRANS.

$e \cdot \dot{e} = -e^\top L^\top (r^* + Le) = -\|Le\|^2 - (Le) \cdot r^*$. This is positive iff $(Le) \cdot r^* < -\|Le\|^2$; taking Le near $-\alpha r^* / \|r^*\|$ (feasible whenever r^* has a nonzero component in range L) the condition holds on an open cone with $0 < \alpha < \|r^*\|$. $\square$

Good detector, bad fixer—from one decomposition. Equation (7) splits the monitored residual into a fixed floor r^* and an error-driven term Le . *Far* from the truth, $\|Le\| \gg \|r^*\|$, so $\|R_h(\hat{u})\| \approx \|Le\|$, and since $\sigma_{\min}(L)\|e\| \leq \|Le\| \leq \sigma_{\max}(L)\|e\|$ the residual norm is a two-sided proxy for the field error (up to $\kappa(L)$)—this is the calibrated *trust signal* (rank correlation up to 0.83, subsection 5.3). *Near* the truth, $\|Le\| \lesssim \|r^*\|$, the floor dominates, and the residual decouples from the error: its minimiser is displaced from u^* by Theorem 1(iii) and a residual-reducing step can grow the error by (iv)—this is the failure of the residual as a correction *objective* (subsection 5.5). Note the divergence in (iv) vanishes when $r^* = 0$: it is driven by the *floor*, not by ill-conditioning, which merely sets the floor’s magnitude in (iii).

The kernel: modes the monitor can neither detect nor fix. L is genuinely non-injective— R_h maps the four field channels (u, v, p, ν_t) to three residual channels (continuity, two momentum), and a uniform pressure offset is an exact null mode (the monitored R_h sees p only through ∇p , and omits the no-slip term that would otherwise pin the gauge). Error components in $\ker L$ —a constant pressure shift, and the residual-blind part of ν_t —are therefore *invisible* to the monitor: it can neither flag them (detector) nor remove them (fixer). The non-trivial kernel is not a defect of the bound but part of the phenomenon: a self-consistent monitor is silent on its own null modes, which is why we report coverage on the pressure channel up to a gauge and never claim the residual certifies ν_t .

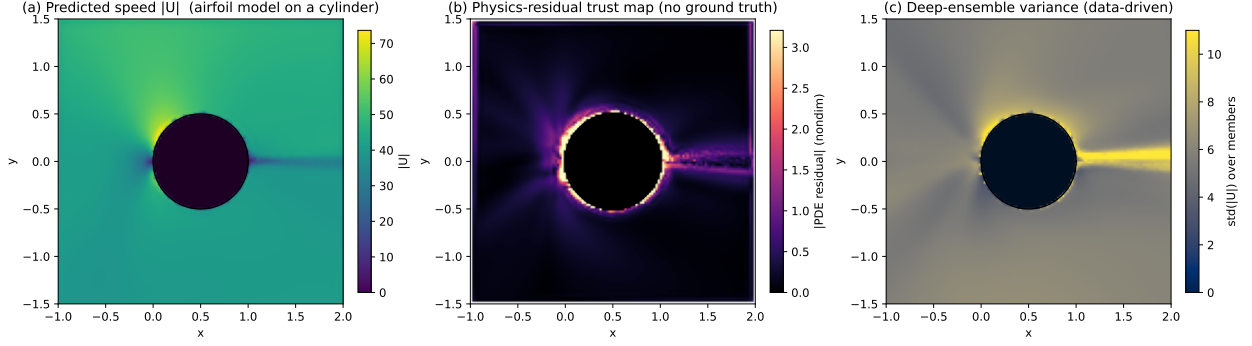


Figure 10: Cross-geometry OOD (qualitative): an airfoil-trained model on a cylinder. (a) predicted speed (visibly broken—the flow is not arrested at the body); (b) the physics-residual map (no ground truth used), elevated field-wide; its bright near-body ring is a generic no-slip-wall feature common to all bodies (a matched in-distribution-airfoil control shows the same near/far ratio), *not* an OOD localiser; (c) deep-ensemble variance (the uncorrected five members, the source of σ), for comparison. The OOD signal is the field-wide residual *magnitude*— $\approx 7\times$ higher than in-distribution airfoils in the far field, away from any wall (subsection 5.4)—not its spatial concentration.

Empirical confirmation. On 200/200 real AirfRANS test cases, the monitored residual of the ground-truth field is substantially nonzero ((H2) holds): $\|r^*\|$ has mean 0.192 (median 0.133), while the uniform-freestream field gives $\|R_h(u_\infty)\| = 0$ in every case—the objective strictly prefers a physically wrong field to the truth (results/certificates/residual_floor_realdata.json). A trained backbone’s over-smoothed prediction (here the dropout-FNO of checkpoints/certificates_deq.pt, distinct from the Transolver headline) sits *below* the truth floor in 160/200 cases, consistent with the loop’s drive away from u^* in Table 5.

Robustness to the boundary term. One might object that the monitored R_h omits the no-slip term, so the negative is merely an artifact of dropping the boundary constraint. We tested this with the *framework’s* own boundary-inclusive residual $\rho_{\text{res}} = \sqrt{r_c^2 + r_x^2 + r_y^2 + r_{bc}^2}$ (the trust-map definition, r_{bc} the proximity-weighted no-slip penalty) along the correction sweep (dropout-FNO+DEQ, $n=10$, same regime as Table 5; results/control/bc_inclusive_sweep.json). The negative *survives*: as the corrector cuts field error (mse_u 3.17 $\rightarrow$ 2.32), the boundary-*inclusive* residual rises (0.145 $\rightarrow$ 0.811) in lockstep with the boundary-excluded one (0.125 $\rightarrow$ 0.807)—along the correction path the prediction and the truth both approximately satisfy no-slip, so r_{bc} stays small and roughly constant and the interior floor dominates. (Empirically the uniform field also keeps a *lower* ρ_{res} than the truth, 0.073 vs 0.187, because r_{bc} is non-dimensionalised by U and contributes only ≈ 0.07 in RMS.) We claim this robustness for the residual the framework computes *as defined*; a no-slip penalty weighted heavily enough to dominate the interior floor is out of scope and untested.

Assumptions, stated plainly. The theorem is about the *monitored discrete* operator R_h —what the loop minimises—not the continuous PDE; (H2) holds here because R_h omits the no-slip closure *and* the 128^2 grid under-resolves the boundary layer and drops the spatially-varying- ν_t term $\nabla \nu_t \cdot \nabla u$. The floor is a *genuine* displacement whenever r^* has a component in range L , which it does— r^* is a discretisation/closure residual, not a pure gauge null mode. The detector leg is a *regime* statement: its tightness is controlled by the conditioning of L on $(\ker L)^\perp$, so we claim a positive rank correlation, not an isometry.

Relation to prior work. Krishnapriyan et al. (2021) tie ill-conditioning to the *optimisation difficulty* of PINN losses, and Wang et al. (2022) analyse spectral bias and loss-term imbalance in PINN *training dynamics*; both concern trainability, not a displacement of the residual minimiser from the truth. The forward-vs-backward error gap $\|e\| \leq \|R_h\|/\sigma_{\min}$ is classical (Trefethen & Bau, 1997), and a-posteriori PINN bounds upper-bound error *by* residual (the benign regime). That a residual blind to certain modes makes the truth a non-minimiser is, in its bare form, elementary (the pressure gauge above). Our contribution

is sharper and threefold: **(a)** the *quantified* operator-specific floor $\|e_\infty\| = \|L^+ r^*\| \leq \|r^*\|/\sigma_{\min}$ for this steady-RANS monitor; **(b)** the *detector/fixer duality*—a single decomposition (7) yielding both the positive trust-signal verdict and the negative objective verdict, including the undetectable-and-uncorrectable kernel modes; and **(c)** the demonstration on a learned neural-operator CFD corrector (AirfRANS), where this is exactly the dissociation our experiments report.

7 Limitations

- **Backbone-dependent correction quality.** The supervised DEQ corrector reduces field MSE by 9–25% on the SOTA Transolver backbone (subsection 5.2) but is flat-to-unhelpful on the weak grid backbone (subsection 5.3). We report both regimes; we do not claim the corrector helps on an arbitrary backbone, only that it helps on a strong one. The trust signal, by contrast, is positive and substantial across all three backbones tested ($\rho = 0.625$ SOTA Transolver, ≈ 0.40 – 0.83 weak grid backbone, 0.851 bare MeshGraphNet GNN). We make no competitiveness claim for the grid backbone (it trails SOTA ~ 4 – $60\times$); the Transolver backbone we run *on* is itself competitive.
- **Grid resolution and near-wall blindness.** A uniform 128^2 Cartesian grid cannot resolve a $\text{Re} \approx 10^6$ boundary layer (sub-cell), so wall quantities are approximate and the absolute MSE values are not comparable to body-fitted solvers. Relatedly, the monitored residual zeroes the solid-adjacent (wall-ring) cells—where the central stencil reaches into the $u = v = 0$ discontinuity and registers spurious values—so the trust signal operates on the bulk and is structurally blind to the immediate near-wall band, precisely where near-wall error concentrates on a high-Re airfoil.
- **Two 2-D datasets; not yet 3-D.** The trust signal is validated on two 2-D datasets—turbulent-RANS AirfRANS airfoils and laminar DeepCFD bluff bodies (subsection 5.4, $\rho = 0.77 \pm 0.12$)—so the phenomenon is not AirfRANS-specific, but all results remain 2-D. Extending the residual verifier to 3-D (AhmedML, Ashton et al., 2024; DrivAerNet++, Elrefaie et al., 2024) requires 3-D residual operators and multi-TB, often unsteady datasets beyond a single-GPU budget; we leave it as genuine future work rather than a near-term claim.
- **Conformal coverage out-of-distribution is empirical, not guaranteed.** The distribution-free guarantee assumes exchangeability, which fails under shift; OOD coverage holds in our experiments only because the uncertainty inflates appropriately (subsection 5.6). Scores are also pooled over spatially-correlated cells, so the effective sample size is below the raw cell count, and the calibration set is small ($n = 100$); a larger-calibration-set robustness check is future work.
- **Residual-as-objective remains a dead end.** Used as a correction objective or acceptance gate, the residual fails: more iterations raise the PDE residual while error falls, and the monotone-residual acceptance test accepts almost no steps (subsection 5.5). The DEQ contraction guarantee is in the correction variable δ , and the acceptance test’s monotone-residual guarantee is real but vacuous as an accuracy mechanism. The correction gains in subsection 5.2 come from the *learned, supervised* correction, not from minimising the residual—and, by the W1 control, not from feeding the corrector the residual either.
- **The corrector’s gain is not attributable to the residual input.** The deployed DEQ corrector ingests the residual as an input channel, but a controlled ablation that zeros that input at train and inference (3 seeds) matches or slightly exceeds it ($\approx 4\%$ on a ≈ 0.11 base; WITH beats NULL on 0/3 seeds; `results/control/w1_capture.json`). We therefore attribute the 9–25% field-MSE improvement to the learned correction architecture, not to residual-conditioning. The residual’s demonstrated value is as a trust signal (*where* the prediction is wrong), not as a correction input (*how* to fix it); the deployed system retains the residual input but does not depend on it for the gain.
- **Adaptive uncertainty on the SOTA model requires a deep ensemble.** On the near-deterministic Transolver backbone, MC-dropout σ at dropout 0.05 is near-degenerate, so although conformal coverage holds (0.902, target 0.90), the quantile q is enormous ($\approx 10^9$) and ECE ≈ 0.31 :

coverage there is delivered by a near-constant band. A deep ensemble of $M=5$ members recovers a genuinely input-adaptive band on this model ($q = 2.35$, ECE 0.074; Table 2), at $5\times$ the training cost. That ensemble certificate comes from a single ensemble training run; the calibration/test split is no longer the caveat (20 re-draws: coverage 0.899 ± 0.009 , subsection 5.6), but independently retrained ensembles and a study of M remain future work.

- **Eddy-viscosity channel near-degenerate.** The ν_t output is effectively unlearned at this scale (per-arm MSE $\approx 5 \times 10^{-8}$); since residuals use $\nu_{\text{eff}} = \nu + \nu_t$, the physics term currently leans on the laminar viscosity. A trainable ν_t target is future work.
- **Classical fallback is a stub interface.** The trust-gated fallback fixes the integration seam and reports what would run; the OpenFOAM/SU2 backends are not implemented and bear on no quantitative claim.
- **Sensitivity sweeps are light.** The iteration and toggle sweeps (subsection 5.5) are single-checkpoint, unseeded sweeps; they illustrate the residual-error divergence robustly in direction but are not multi-seed.

8 Conclusion

We set out to use the steady-RANS physics residual of a neural-CFD surrogate, and we found that it plays **two distinct roles with two distinct verdicts**. As a *trust signal* it is a reliable, backbone-robust (three architecturally-distinct backbones) case-level detector of error—it tells you *where* a prediction is wrong: $\rho = 0.625\pm 0.019$ on a SOTA Transolver backbone, $\approx 0.40\text{--}0.83$ on a weak grid backbone, 0.851 ± 0.058 on a bare message-passing GNN (MeshGraphNet), and—under distribution shift on the weak backbone—regime-invariant (≈ 0.74 , rescuing an *aoa* collapse from 0.31 to 0.75) where the one-shot signal collapses. As a *correction objective or acceptance gate* it fails: more iterations raise the residual while error falls, and the monotone-residual gate accepts almost nothing—minimising the residual is not minimising the error. Alongside the trust signal we deploy a learned, supervised learned correction that works on a strong backbone: the DEQ corrector cuts volume field MSE by 9–25% on all 3 seeds of the SOTA Transolver model (gate-verified), with force ranking held; a controlled ablation attributes that gain to the learned correction rather than to the residual input (the residual tells you *where* to look, not *how* to fix). On the strength of the detector result we package a backbone-robust conformal trust layer that holds target coverage both in-distribution (0.91/0.93/0.94 at 0.90, weak backbone) and on the SOTA backbone (0.902), and empirically retains coverage out-of-distribution via uncertainty inflation; pairing it with a deep-ensemble σ makes the band input-adaptive even on the near-deterministic SOTA model ($q = 2.35$, ECE 0.074), where MC-dropout alone gives a near-constant one. Two honest caveats accompany the corrector: correction quality is backbone-dependent (flat on the weak backbone), and a controlled ablation attributes its gain to the learned correction rather than to the residual input it ingests. The durable contribution is the self-auditing, backbone-robust trust layer and the learned, supervised corrector it accompanies—demonstrated on a competitive surrogate, reproducible end-to-end from the committed harness.

Code and data availability

The complete source, training/evaluation harnesses, and per-claim reproduction map (docs/REPRODUCE.md) are released as the open-source package `neuroforge-cfd` (<https://github.com/ali-kin4/neuroforge-cfd>; archived at Zenodo, DOI 10.5281/zenodo.21277928). Every headline number maps to a committed script and result file, and a top-level manifest (results/MANIFEST.json) records seeds, environment, and SHA-256 hashes for hash-drift detection. The two benchmarks are public: AirFRANS (Bonnet et al., 2022) and DeepCFD (Ribeiro et al., 2020). The package is CPU-first and runs end-to-end with zero downloads via a synthetic data generator; the GPU runs (backbone training) are documented with exact commands and expected outputs.

CRediT author contribution statement

Ali Jabbary: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Kasra Ghanavati:** Validation, Writing – review & editing.

Declaration of competing interests

The authors declare no competing financial or personal interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist in drafting and editing the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The use of AI assistance in developing the research software is described in the Implementation section.

References

- Neil Ashton, Danielle Maddix, Samuel Gundry, and Parisa Shabestari. AhmedML: High-fidelity computational fluid dynamics dataset for incompressible, low-speed bluff body aerodynamics. *arXiv preprint arXiv:2407.20801*, 2024.
- Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. arXiv:1909.01377.
- Florent Bonnet, Jocelyn Ahmed Mazari, Paola Cinnella, and Patrick Gallinari. AirFRANS: High fidelity computational fluid dynamics dataset for approximating reynolds-averaged navier-stokes solutions. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2022. arXiv:2212.07564.
- Mohamed Elrefaie, Florin Morar, Angela Dai, and Faez Ahmed. DrivAerNet++: A large-scale multimodal car dataset with computational fluid dynamics simulations and deep learning benchmarks. *arXiv preprint arXiv:2406.09624*, 2024.
- Samy Wu Fung, Howard Heaton, Qiuwei Li, Daniel McKenzie, Stanley Osher, and Wotao Yin. JFB: Jacobian-free backpropagation for implicit networks. In *AAAI Conference on Artificial Intelligence*, 2022. arXiv:2103.12803.
- Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, 2016. arXiv:1506.02142.
- Vignesh Gopakumar, Ander Gray, Lorenzo Zanisi, Timothy Nunn, Daniel Giles, Matt J. Kusner, Stanislas Pamela, and Marc Peter Deisenroth. Calibrated physics-informed uncertainty quantification. *arXiv preprint arXiv:2502.04406*, 2025.
- Jun-Ting Hsieh, Shengjia Zhao, Stephan Eismann, Lucia Mirabella, and Stefano Ermon. Learning neural PDE solvers with convergence guarantees. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1906.01200.
- Prashant K. Jha. Residual-based error corrector operator to enhance accuracy and reliability of neural operator surrogates of nonlinear variational boundary-value problems. *Computer Methods in Applied Mechanics and Engineering*, 419:116595, 2024. doi: 10.1016/j.cma.2023.116595. arXiv:2306.12047.

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- Chundong Jia, Chao Xia, Alexey Vdovin, Qing Jia, Simone Sebben, and Zhigang Yang. Multi-granularity conformal prediction for reliable neural-operator automotive aerodynamic surrogates. *arXiv preprint arXiv:2607.17297*, 2026.
- Aditi S. Krishnapriyan, Amir Gholami, Shandian Zhe, Robert M. Kirby, and Michael W. Mahoney. Characterizing possible failure modes in physics-informed neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. arXiv:1612.01474.
- Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2010.08895.
- Zongyi Li, Daniel Z. Huang, Burigede Liu, and Anima Anandkumar. Fourier neural operator with learned deformations for pdes on general geometries. *arXiv preprint arXiv:2207.05209*, 2022.
- Phillip Lippe, Bastiaan S. Veeling, Paris Perdikaris, Richard E. Turner, and Johannes Brandstetter. PDE-refiner: Achieving accurate long rollouts with neural PDE solvers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2308.05732.
- Huakun Luo, Haixu Wu, Hang Zhou, Lanxiang Xing, Yichen Di, Jianmin Wang, and Mingsheng Long. Transolver++: An accurate neural solver for PDEs on million-scale geometries. *arXiv preprint arXiv:2502.02414*, 2025.
- Ziqi Ma, Kamyar Azizzadenesheli, and Anima Anandkumar. Calibrated uncertainty quantification for operator learning via conformal prediction. *Transactions on Machine Learning Research (TMLR)*, 2024. arXiv:2402.01960.
- Tanya Marwah, Ashwini Pokle, J. Zico Kolter, Zachary C. Lipton, Jianfeng Lu, and Andrej Risteski. Deep equilibrium based neural operators for steady-state PDEs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2312.00234.
- Amartya Mukherjee, Maxwell Fitzsimmons, David C. Del Rey Fernández, and Jun Liu. Rigorous error certification for neural PDE solvers: From empirical residuals to solution guarantees. *arXiv preprint arXiv:2603.19165*, 2026.
- Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- Rishikesh Ranade, Mohammad Amin Nabian, Kaustubh Tangsali, Alexey Kamenev, Oliver Hennigh, Ram Cherukuri, and Sanjay Choudhry. DoMINO: A decomposable multi-scale iterative neural operator for modeling large scale engineering simulations. *arXiv preprint arXiv:2501.13350*, 2025. NVIDIA.
- Mateus Dias Ribeiro, Abdul Rehman, Sheraz Ahmed, and Andreas Dengel. DeepCFD: Efficient steady-state laminar flow approximation with deep convolutional neural networks. *arXiv preprint arXiv:2004.08826*, 2020.
- Rajyasri Roy, Dibyajyoti Nayak, and Somdatta Goswami. ANCHOR: Error-controlled adaptive numerical correction for neural operator time marching. *arXiv preprint arXiv:2512.19643*, 2025.
- Haoze Song, Zhihao Li, Mengyi Deng, Xin Li, Duyi Pan, Zhilu Lai, and Wei Wang. Structure-aware epistemic uncertainty quantification for neural operator PDE surrogates. *arXiv preprint arXiv:2603.11052*, 2026.
- Lloyd N. Trefethen and David Bau. *Numerical Linear Algebra*. SIAM, 1997.

Sifan Wang, Xinling Yu, and Paris Perdikaris. When and why PINNs fail to train: A neural tangent kernel perspective. *Journal of Computational Physics*, 449:110768, 2022.

Ezra Winston and J. Zico Kolter. Monotone operator equilibrium networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. arXiv:2006.08591.

Haixu Wu, Huakun Luo, Haowen Wang, Jianmin Wang, and Mingsheng Long. Transolver: A fast transformer solver for PDEs on general geometries. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.02366.